
Exploring the effect of pressure on the optoelectronic, mechanical, and thermoelectric activities of double perovskite $\text{Ba}_2\text{NbRhO}_6$

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Abstract

Recent energy deficits have significantly intensified the global demand for renewable energy sources. For the utilization of these energy sources, materials capable of absorbing light across the visible and ultraviolet (UV) spectrum are required. In this study, the electronic, optical, mechanical, and thermoelectronic properties of $\text{Ba}_2\text{NbRhO}_6$ double perovskite have been investigated utilizing density functional theory (DFT) under hydrostatic pressures ranging from 0 to 100 GPa. The structural stability of the material is confirmed by a Goldsmith's tolerance factor of 0.91 and a negative formation energy of -3.48 eV/atom. The analysis of electronic properties reveals that the material has a semiconducting nature with an indirect band gap of 0.861 eV at ambient condition, which increases progressively with applied pressure, reaching 1.860 eV at 100 GPa. Optical analysis has been conducted through the calculation of dielectric function, absorption coefficient, extinction coefficient, optical conductivity, reflectivity, refractive index and energy loss function. The optical absorption edges move toward the lower energy region (red shift), and both absorption and conductivity increase with applied pressure, enhancing the material's suitability for optoelectronic applications in the ultraviolet region. The mechanical stability of $\text{Ba}_2\text{NbRhO}_6$ is confirmed by the Born stability criteria, and the material exhibits ductile behavior at 0–100 GPa. The thermoelectric properties are investigated using the BoltzTraP code to estimate the electrical and thermal conductivities, Seebeck coefficient, power factor, and figure of merit under different hydrostatic pressures. Additionally, low thermal conductivity and high figure of merit value suggest that the studied material is a promising candidate for application in thermoelectric devices, such as thermoelectric generators and coolers.

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Chapter 1

Introduction

In recent years, global energy consumption has increased steadily due to urban migration and rising demand for digital devices, resulting in an energy crisis. Due to the limitations of conventional energy resources and their adverse effects on both the economy and the environment, it has become crucial to explore renewable energy sources in pursuit of sustainable development [1]. In the present situation, scientists and engineers are focusing their attention towards various energy conversion devices, such as solar cells and thermoelectric generators, which offer a sustainable and eco-friendly alternative to fossil fuels [2–4]. For this purpose, oxide-based double perovskite materials are regarded as a crucial class of materials for energy conversion and storage applications [5,6]. These materials follow the general formula $A_2BB'O_6$, where A stands for an alkaline or rare earth ion, B and B' are two different transition metals, and O is an anion [7,8]. Double perovskites have become highly influential in the field of energy harvesting, including applications in solar cells, light-emitting diodes, photovoltaic cells, sensors, fuel cells, lasers, thermoelectrics, X-ray detectors, photodetectors, and photocatalysis [9–15]. In solar cells, charge-carrier recombination is a critical factor, while in photocatalysis, the photoexcited electrons play a central role in driving the process. Double perovskite oxide materials have gained significant attention in the research community for photovoltaic and thermoelectric

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applications, with their efficiency increasing steadily from 3.8% in 2009 to 25.7% at present [16, 17].

Recently, many researchers have been investigating oxide-based double perovskite materials extensively, both at ambient condition and different hydrostatic pressures. The study of Hansraj et al. [18] on the Ba-based direct band gap semiconducting material Ba_2SbXO_6 ($X = \text{Nb}, \text{Ta}$). They found that $\text{Ba}_2\text{SbNbO}_6$ and $\text{Ba}_2\text{SbTaO}_6$ exhibit mechanically stable and ductile nature, also show great potential for photovoltaic and thermoelectric applications because of their high absorption spectra in UV region and good figure of merit (ZT). The thermoelectric properties of $\text{Ba}_2\text{FeMoO}_6$ was investigated by Sahnoun et al. [19] and confirmed a highly effective thermoelectric material due to its high ZT value near unity at low temperatures. In 2024, Belhachi et al. [20] explored the physical properties of $\text{Ba}_2\text{NbRhO}_6$, focusing its potential for applications in photovoltaic and thermoelectric devices. Shabir et al. [21] have performed a theoretical analysis of the oxide double perovskite $\text{Ba}_2\text{LaNbO}_6$ and observed promising thermoelectric properties, making it suitable for renewable energy applications. Qi Dai et al. [22] investigated the structural, optoelectronic, and mechanical properties of $\text{Ba}_2\text{ScAsO}_6$ and $\text{Ba}_2\text{ScAsO}_6$ using density functional theory (DFT) approach and reported that these materials exhibit indirect bandgap of 3.829 eV and 4.796 eV, respectively. In addition, they demonstrated effective optical absorption of $\text{Ba}_2\text{ScAsO}_6$ and $\text{Ba}_2\text{ScAsO}_6$ within the UV spectrum that make them competent in photovoltaic applications.

The effect of pressure on electronic, magnetic, thermodynamic, and thermoelectric properties of cubic double perovskite Ba_2MTaO_6 ($M = \text{Mn}, \text{Cr}$) were reported by Muskan et al. [23] and observed that both materials exhibit half-metallic nature as the pressure increases from 0 to 25 GPa. Yanju Wang et al. [24] explored the pressure-dependent structural behavior and electronic properties of double perovskite $\text{Ba}_2\text{SmSbO}_6$. Their study provide essential insights into the structure property modulation of $\text{Ba}_2\text{SmSbO}_6$ under high pressure and highlights its potential for use in pressure sensor applications. Lokman Ali et al. [25] investigated pressure-induced physical properties of $\text{La}_2\text{NiMnO}_6$ and their research confirms that with

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rising pressure, the material transition occurs from brittle to ductile behavior, which improved its stiffness and flexibility. Sholagberu et al. [26] studied the impact of pressure on the optoelectronic and mechanical properties of the double perovskite $\text{Cs}_2\text{AgInCl}_6$ and their findings suggest that the material could be a promising candidate for the absorbing layer in thin-film solar cells due to its direct bandgap and tunable optoelectronic properties under pressure. Fahim et al. have reported first-principles calculations of $\text{Rb}_2\text{NaScCl}_6$ double perovskite implying different hydrostatic pressures, revealing high absorption in the UV region and a high figure of merit of 0.81 at room temperature, indicating its potential for optoelectronic and thermoelectric applications [27].

Nowadays, influence of pressure on perovskite compounds is a subject of intense research, revealing intriguing effects on both the structural and electronic characteristics of these materials [28–31]. Understanding the pressure-dependent behavior of perovskites is crucial for tailoring their properties to specific applications, ranging from optoelectronic devices to energy storage systems [32–34]. Under pressure, changes in the lattice parameters can be observed, impacting the arrangement of atoms within the crystal lattice [35]. Notably, alterations in the electronic band structure occur, influencing the electrical conductivity and optical properties of perovskites. The application of pressure may also induce phase transitions, leading to modifications in symmetry and overall crystal structure [36]. Researchers are actively exploring how these pressure-induced changes can enhance the functionalities of perovskite materials, opening up possibilities for improved performance in applications such as optoelectronic devices and energy storage systems.

We perform the first principle calculation method using Density Functional Theory (DFT) as implemented in WIEN2k code. DFT is a computational quantum mechanical method used to study the electronic structure and properties of solids. It simplifies calculations by focusing on electron density rather than directly solving complex equations, making it a valuable tool for predicting material properties, simulating chemical reactions, and advancing research in materials science and nanotechnology [37]. WIEN2k, a powerful software package developed by Peter Blaha

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and his team, utilizes the Full Potential Linearized Augmented Plane Wave (FP-LAPW) method within the framework of DFT [38]. Renowned for its accuracy, WIEN2k is versatile in studying various materials, from metals to insulators and magnetic substances. It can handle a wide range of crystal structures and is particularly useful for investigating complex and challenging materials. BoltzTraP is employed to study the electronic transport properties and explore possible thermoelectric applications. We investigate the studied material under both ambient and different hydrostatic pressures because material behavior changes with application of a strain or to model interactions between a substrate and molecule.

Previous studies have revealed the promising optoelectronic and thermoelectronic properties of $\text{Ba}_2\text{NbRhO}_6$ under ambient pressure. Motivated by these findings, we aim to investigate its physical properties under pressurized conditions. To the best of our knowledge, this is the first study to report the optoelectronic, mechanical, and thermoelectric properties of the material under hydrostatic pressure. The current study explores the intriguing properties of $\text{Ba}_2\text{NbRhO}_6$ double perovskite under pressure. The purpose of this study is to enhance the material's optoelectronic and thermoelectric properties in the UV regions, we systematically analyze the physical properties of $\text{Ba}_2\text{NbRhO}_6$ under varying hydrostatic pressures using the full-potential linearized augmented plane wave (FP-LAPW) method within the framework of DFT.

The arrangement of this thesis is such that in Chapter 2, we discuss basic quantum mechanics as the base of density functional theory, starting from Schrödinger equation, and also discuss the Born-Oppenheimer approximation, and the Hartree-Fock method along with its limitations. In chapter 3 we discuss the theoretical density functional theory, including the electron density, Thomas Fermi model, the Hohenberg Khon theorems, Kohn Sham equation with the corresponding flowchart, and the exchange–correlation functionals. In chapter 4, we discuss the results of the compound under both ambient and pressurized conditions. In last chapter, we make a conclusion about the compound and its applicability as a practical device.

Basic Quantum Mechanics

2.1 Schrödinger Equation

The Schrödinger equation is a fundamental equation in quantum mechanics that describes the quantum state of a physical system. It is crucial for understanding the behavior of particles at the atomic and subatomic levels, such as electrons, photons, and other quantum objects. First formulated by Erwin Schrödinger in 1925 [39], this equation provides a mathematical framework for predicting the properties and behavior of quantum systems. The Schrödinger equation can be time-dependent and time-independent. The mathematical form of time-independent Schrödinger equation is

$$\hat{H}\psi(\vec{r}) = E\psi(\vec{r}). \quad (2.1)$$

where, $\hat{H}$ is the Hamiltonian, ψ is the wave function and E is the energy eigenvalue. This equation is crucial in finding the stationary states of quantum systems. The Hamiltonian $\hat{H}$, represents the total energy operator of the system and is typically composed of two parts: the kinetic energy operator, $\hat{T} = -\frac{\hbar^2}{2m}\vec{\nabla}^2$ and the potential energy operator, $\hat{V} = V(\vec{r})$. The kinetic energy operator describes the motion of particles, while the potential energy operator accounts for the forces acting on them due to their positions within the system. For a single particle, it is expressed by the

following equation:

$$\hat{H} = \hat{T} + \hat{V} = -\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r}). \quad (2.2)$$

this equation leads to the time-independent single-particle Schrödinger equation

$$E\psi(\vec{r}) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r})\right]\psi(\vec{r}). \quad (2.3)$$

For N particles in three dimensions, the Hamiltonian is

$$\hat{H} = \sum_{i=1}^N \frac{\hat{p}_i^2}{2m_i} + V(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N). \quad (2.4)$$

The corresponding Schrödinger equation is

$$E\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) = \left[\sum_{i=1}^N \frac{\hat{p}_i^2}{2m_i} + V(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)\right]\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N). \quad (2.5)$$

2.1.1 The Wave Function

The wave function, $\psi(\vec{r})$, is a fundamental concept in quantum mechanics. It encapsulates all the information about the quantum state of a system, including the likelihood of finding particles in particular positions $\vec{r}$ and with specific properties. In essence, ψ encodes the physical behavior of the system and allows us to calculate observable quantities, such as energy, by solving the Schrödinger equation. The wave function is generally a complex function that depends on the position of particles in the system. While ψ itself does not have direct physical meaning, it is crucial for calculating observable properties of particles. The square of the wave function, $|\psi|^2$, represents the probability distribution for the positions of particles, offering critical insights into their behavior within the quantum system.

$$|\psi|^2 = \psi^* \psi. \quad (2.6)$$

The wave function ψ must be finite everywhere. If ψ is finite for a particular point, that means an infinite larger probability of finding the particles at that point. The wave function must be single valued. If it has more than one value at any point,

that means more than one value of probability of finding the particle at that point which is obviously improbable. The wave function must be continuous and have a continuous first derivative everywhere and its must be normalized. For the sake of simplicity the discussion is restricted to the time-independent wave function. A question always arising with physical quantities is about possible interpretations as well as observations. The Born probability interpretation of the wave function, which is a major principle of the Copenhagen interpretation of quantum mechanics, provides a physical interpretation for the square of the wave function as a probability density [40, 41].

$$P = |\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)|^2 d\vec{r}_1 d\vec{r}_2, \dots, d\vec{r}_N. \quad (2.7)$$

Equation (2.7) describes the probability that particles 1,2,...,N are located simultaneously in the corresponding volume element $d\vec{r}_1, d\vec{r}_2, \dots, d\vec{r}_N$. What happens if the positions of two particles are exchanged, must be considered as well. Following merely logical reasoning [42], the overall probability density cannot depend on such an exchange, i.e.

$$|\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_i, \vec{r}_j, \dots, \vec{r}_N)|^2 = |\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_j, \vec{r}_i, \dots, \vec{r}_N)|^2. \quad (2.8)$$

There are only two possibilities for the behavior of the wave function during a particle interchange. The first one is a symmetrical wave function, which does not change due to such an interchange. This corresponds to bosons (particles with integer or zero spin). The other possibility is an antisymmetrical wave function, where an exchange of two particles causes a sign change, which corresponds to fermions interest, which are fermions. The antisymmetric fermion wave function leads to the Pauli principle, which states that no two electrons can occupy the same state, whereas state means the orbital and spin parts of the wave function [43, 44] (the term spin coordinates will be discussed later in more detail). The antisymmetry principle can be seen as the quantum mechanical formalization of Pauli's theoretical ideas in the description of spectra (e.g. alkaline doublets). Another consequence of the probability of finding a particle in a volume element, setting the full range of coordinates as volume element must result in a probability of one, i.e. all particles must be found somewhere in

space. This corresponds to the normalization condition for the wave function.

$$\int d\vec{r}_1 \int d\vec{r}_2 \dots \int d\vec{r}_N |\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)|^2 = 1. \quad (2.9)$$

Equation (2.9) also gives insight on the requirements a wave function must fulfill in order to be physical acceptable. Wave functions must be continuous over the full spatial range and square-integratable [45]. Calculating the expectation value of the relevant observable for that wave function. For an observable $O(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$, this can be written as

$$O = \langle O \rangle = \int d\vec{r}_1 \int d\vec{r}_2 \dots \int d\vec{r}_N \psi^* (\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \langle \hat{O} \rangle \psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N). \quad (2.10)$$

2.2 The Many-Body Problem

The many-body problem in quantum mechanics addresses the challenge of describing systems comprising multiple interacting atoms. Unlike single-atom systems, which can often be analyzed using analytical solutions, many-body systems involve intricate interactions that lead to emergent phenomena, such as correlations and collective behaviors. The Hamiltonian of a many-body system consisting of nuclei and electrons can be written as,

$$\hat{H} = \hat{T}_n + \hat{T}_e + \hat{V}_{nn} + \hat{V}_{ee} + \hat{V}_{ne}. \quad (2.11)$$

Where,

$\hat{T}_n$: the kinetic energy of the nuclei,

$\hat{T}_e$: the kinetic energy of the electrons.

$\hat{V}_{nn}$: the nucleus-nucleus Coulomb repulsion,

$\hat{V}_{ee}$: the electron-electron Coulomb repulsion,

$\hat{V}_{ne}$: the attractive interaction between nuclei and electrons.

These interaction terms make solving the Schrödinger equation for large systems of atoms exceedingly complex. The Coulomb interactions $\hat{V}_{nn}$, $\hat{V}_{ee}$ and $\hat{V}_{ne}$ describe the

forces between particles, which must be accounted for when determining the total energy of the system. As the number of atoms increases, the number of interactions grows rapidly, making direct analytical or numerical solutions impractical [46]. With the explicit forms of the kinetic energies and interaction terms, the many-body Hamiltonian for a system consisting of nuclei and electrons becomes [47]:

$$\hat{H}_{tot} = - \sum_I \frac{\hbar^2}{2M_I} \nabla_{\vec{R}_I}^2 - \sum_i \frac{\hbar^2}{2m_e} \nabla_{\vec{r}_i}^2 + \frac{1}{2} \sum_{\substack{I,J \\ I \neq J}} \frac{Z_I Z_J e^2}{|\vec{R}_I - \vec{R}_J|} \quad (2.12)$$

$$+ \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} \frac{e^2}{|\vec{r}_i - \vec{r}_j|} - \sum_{I,i} \frac{Z_I e^2}{|\vec{R}_I - \vec{r}_i|}. \quad (2.13)$$

Where the indexes I, J run on nuclei i and j on electrons, $\vec{R}_I$ and M_I are positions and masses of the nuclei, $\vec{r}_i$ and m_e of the electrons, Z_I the atomic number of nucleus I . The first term is the kinetic energy of the nuclei, the second term is the kinetic energy of the electrons, the third term is the potential energy of the nucleus–nucleus Coulomb interaction, the fourth term is the potential energy of electron–electron Coulomb interaction, and the last term represents the potential energy of nucleus–electron Coulomb interaction.

Solving the many–body Hamiltonian presents several significant challenges in quantum mechanics, particularly in systems involving both electrons and nuclei. The high dimensionality of the problem is a major hurdle; for a system with N electrons and M nuclei, the wave function depends on $3(N + M)$ variables, making analytical solutions increasingly intractable as N and M increase. Furthermore, the Coulomb interactions between electrons introduce complex correlations, complicating the treatment of each electron as an independent particle. Furthermore, since electrons are fermions, they must adhere to the Pauli exclusion principle, requiring that the many-body wave function is antisymmetric under the exchange of particles, which adds another layer of complexity. The potential energy terms in the Hamiltonian, which encompass nucleus-nucleus and nucleus-electron interactions, further contribute to the difficulty. Although analytical solutions are only feasible for very simple systems, such as the hydrogen atom, most many-body systems with multiple

electrons and nuclei do not permit exact solutions.

2.2.1 Born-Oppenheimer (BO) Approximation

Given the complexities associated with the many-body Hamiltonian, particularly in systems containing both electrons and nuclei, it becomes essential to simplify the problem for practical analysis. One of the most widely used approaches is the Born-Oppenheimer approximation, which exploits the significant difference in mass and, consequently, the motion between electrons and nuclei. Since nuclei are much more massive than electrons, their motion can be considered relatively slow compared to the fast-moving electrons [48]. This allows us to treat the nuclei as fixed in space while solving for the electronic wave functions. By separating the total wave function into electronic and nuclear components, the Born-Oppenheimer approximation reduces the complexity of the many-body Schrödinger equation. The Hamiltonian of a many-body system consisting of nuclei and electrons can be written as

Under this approximation, the total Hamiltonian can be expressed as a sum of two parts: the electronic Hamiltonian $\hat{H}_e$, which describes the motion of electrons in the field of fixed nuclei, and the nuclear Hamiltonian $\hat{H}_n$, which accounts for the motion of the nuclei interacting with each other. The total Hamiltonian $\hat{H}$ is then given by [44]:

$$\hat{H} = \hat{H}_e + \hat{H}_n \quad (2.14)$$

where

$$\hat{H}_e = - \sum_i \frac{\hbar^2}{2m_e} \nabla_{\vec{r}_i}^2 + \frac{1}{2} \sum_{i,j} \frac{e^2}{|\vec{r}_i - \vec{r}_j|} - \sum_{I,i} \frac{Z_I e^2}{|\vec{R}_I - \vec{r}_i|}. \quad (2.15)$$

and

$$\hat{H}_n = \sum_I \frac{\hbar^2}{2M_I} \nabla_{\vec{R}_I}^2 + \frac{1}{2} \sum_{I,J} \frac{Z_I Z_J e^2}{|\vec{R}_I - \vec{R}_J|}. \quad (2.16)$$

This simplification allows for easier calculations since the focus is primarily on the electronic structure. In many cases, especially in density functional theory, the nuclei are often treated as fixed due to their relatively large mass compared to electrons.

This approximation streamlines the calculations by reducing the complexity of the system to that of the electrons interacting in a static potential created by the fixed nuclei. However, if nuclear motion needs to be included, the Schrödinger equation for the nuclei can be solved, treating them as quantum mechanical entities. The total energy E_{total} of the system is then obtained as the sum of the electronic energy and the nuclear energy.

This dual approach captures the intricate interplay between electronic and nuclear dynamics, providing a comprehensive view of the system's energy by accounting for both contributions. Once the electronic Hamiltonian $\hat{H}_e$ is defined, the electronic Schrödinger equation can be expressed as:

$$\hat{H}_e\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) = E\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \quad (2.17)$$

In this equation, $\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$ is the many-body wave function, which depends on the positions of all N electrons, and E represents the total energy of the system. Solving this many-body electronic Schrödinger equation poses several challenges. The wave function ψ is a complex function of all electron coordinates, making it difficult to handle analytically, especially as N increases. Additionally, capturing the correlation between electrons, particularly in systems with strong electron-electron interactions, adds to the complexity. Using all approximations introduced up to now, it is possible to calculate a problem similar to H_2^+ , a single ionized hydrogen molecule. To get results for larger systems, further approximations have to be made.

2.3 Hartree-Fock (HF) Approximation

In order to find a suitable strategy to approximate the analytically not accessible solutions of many-body problems, a very useful tool is variational calculus, similar to the least action principle of classical mechanics. By the use of variational calculus, the ground state wave function ψ_o , which corresponds to the lowest energy of the system E_o can be approached.

Hence, for now only the electronic Schrödinger equation is of interest, therefore,

in the following sections we set $\hat{H} \equiv \hat{H}_e$, $\hat{E} \equiv \hat{E}_e$, and so on. Observables in quantum mechanics are calculated as the expectation values of operators [48, 49]. The energy as observable corresponds to the Hamiltonian operator, therefore the energy corresponding to a general Hamiltonian can be calculated as

$$E = \langle \hat{H} \rangle = \int d\vec{r}_1 \int \vec{r}_2, \dots \int d\vec{r}_N \psi^*(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \hat{H} \psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \quad (2.18)$$

The Hartree-Fock technique is based on the principle that the energy obtained by any (normalized) trial wave function other than the actual ground state wave function is always an upper bound, i.e. higher than the actual ground state energy. If the trial function happens to be the desired ground state wave function, the energies are equal.

$$E_{trial} \geq E_o \quad (2.19)$$

$$E_{trial} = \int d\vec{r}_1 \int \vec{r}_2, \dots \int d\vec{r}_N \psi_{trial}^*(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \hat{H} \psi_{trial}(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \quad (2.20)$$

and

$$E_o = \int d\vec{r}_1 \int \vec{r}_2, \dots \int d\vec{r}_N \psi_o^*(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \hat{H} \psi_o(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N) \quad (2.21)$$

For a detailed description of this notation, the reader is referred to the original publication [50]. In that notation, equation (2.18) to (2.21) are expressed as

$$\langle \psi_{trial} | \hat{H} | \psi_{trial} \rangle = E_{trial} \geq E_o = \langle \psi_o | \hat{H} | \psi_o \rangle \quad (2.22)$$

Proof : The eigenfunctions ψ_i of the Hamiltonian $\hat{H}$ (each corresponding to an energy eigenvalue E_i form a complete basis set, therefore any normalized trial wave function ψ_{trial} can be expressed as linear combination of those eigenfunctions.

$$\psi_{trial} = \sum_i \lambda_i \psi_i \quad (2.23)$$

The assumption is made that the eigenfunctions are orthogonal and normalized.

Hence it is requested that the trial wave function is normalized, it follows that

$$\langle \psi_{trial} | \hat{H} | \psi_{trial} \rangle = 1 = \langle \sum_i \lambda_i \psi_i | \sum_j \lambda_j \psi_j \rangle = \sum_i \sum_j \lambda_i^* \lambda_j \langle \psi_i | \psi_j \rangle = \sum_j |\lambda_j|^2 \quad (2.24)$$

On the other hand, following (2.22) and (2.24)

$$E_{trial} = \langle \psi_{trial} | \hat{H} | \psi_{trial} \rangle = \langle \sum_i \lambda_i \psi_i | \hat{H} | \sum_j \lambda_j \psi_j \rangle = \sum_j E_j |\lambda_j|^2 \quad (2.25)$$

Together with the fact that the ground state energy E_o is per definition the lowest possible energy, and therefore has the smallest eigenvalue ($E_o \leq E_i$), it is found that

$$E_{trial} = \sum_j |E_j \lambda_j|^2 \geq E_o \sum_j |\lambda_j|^2 \quad (2.26)$$

For the minimal energy value while applied on all allowed N -electron wave functions will always provide the ground-state wave function (or wave functions, in case of a degenerate ground state where more than one wave function provides the minimum energy). Expressed in terms of functional calculus, where $\psi \rightarrow N$ addresses all allowed N -electron wave functions,

$$E_o = \min_{\psi \rightarrow N} E[\psi] = \min_{\psi \rightarrow N} \langle \psi | \hat{H} | \psi \rangle = \min_{\psi \rightarrow N} \langle \psi | \hat{T} + \hat{V} + \hat{U} | \psi \rangle \quad (2.27)$$

Due to the vast number of alternative wave functions, on the one hand, the constrained processing capacity and time, the solution for the N electron systems is almost unachievable. As in the restricted Hartree–Fock approach, it is possible to limit the search to a manageable subset of wave functions. The search is restricted to the antisymmetric product of N electron wave functions that approximates N wave function of this type is called Slater–determinant.

$$\phi_o \approx \phi_{SD} = (N!)^{-\frac{1}{2}} \begin{bmatrix} \chi_1(\vec{x}_1) & \chi_2(\vec{x}_1) & \dots & \chi_N(\vec{x}_1) \\ \chi_1(\vec{x}_2) & \chi_2(\vec{x}_2) & \dots & \chi_N(\vec{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_1(\vec{x}_N) & \chi_2(\vec{x}_N) & \dots & \chi_N(\vec{x}_N) \end{bmatrix} \quad (2.28)$$

It is important to notice that the spin-orbitals $\chi_i(x_i)$ are not only depending on spatial coordinates but also on a spin coordinate which is introduced by a spin function [51], $x_i = \vec{r}_i, s$. Returning to the variational principle and equation (2.28), the ground state energy approximated by a single slater determinant becomes

$$E_0 = \min_{\phi_{SD} \rightarrow N} E[\phi_{SD}] = \min_{\phi_{SD} \rightarrow N} \langle \phi_{SD} | \hat{H} | \phi_{SD} \rangle = \min_{\phi_{SD} \rightarrow N} \langle \phi_{SD} | \hat{T} + \hat{V} + \hat{U} | \phi_{SD} \rangle \quad (2.29)$$

A general expression for the Hartree-Fock energy is obtained using of the Slater determinant as a trial function.

$$E_{HF} = \langle \phi_{SD} | \hat{H} | \phi_{SD} \rangle = \sum_i^N H_i + \frac{1}{2} \sum_{i,j=1}^N (J_{ij} - K_{ij}). \quad (2.30)$$

Where, the first term corresponds to the kinetic energy and the nucleus-electron interactions. So, the single particle contribution of the Hamiltonian is written as,

$$H_i = \int \psi_i^*(\vec{x}) \left[-\frac{1}{2} \nabla^2 + V(\vec{x}) \right] \psi_i(\vec{x}) d\vec{x}. \quad (2.31)$$

And the last term of equation(2.30) correspond electron-electron interactions. They are called Coulomb (J_{ij}) and exchange integral (K_{ij}) [52]. We can write this term in the following way,

$$J_{ij} = \int \int \psi_i(\vec{x}_1) \psi_i^*(\vec{x}_1) \frac{1}{r_{12}} \psi_j^*(\vec{x}_2) \psi_j(\vec{x}_2) d\vec{x}_1 d\vec{x}_2. \quad (2.32)$$

$$K_{ij} = \int \int \psi_i^*(\vec{x}_1) \psi_j(\vec{x}_1) \frac{1}{r_{12}} \psi_i(\vec{x}_2) \psi_j^*(\vec{x}_2) d\vec{x}_1 d\vec{x}_2. \quad (2.33)$$

These integrals are all real, and $J_{ij} \geq K_{ij} \geq 0$. We have the important equation

$$J_{ii} = K_{ii} \quad (2.34)$$

This is the reason the double summation in the equation that include $i = j$ terms. Minimization of equation subject to the orthonormalization conditions,

$$\psi_i^*(x) \psi_j(x) dx = \delta_{ij}. \quad (2.35)$$

gives the Hartree–Fock differential equation

$$\hat{F}\psi_i(x) = \sum_{j=1}^N \epsilon_{ij} \psi_j(x). \quad (2.36)$$

Where,

$$\hat{F} = \frac{1}{2} \nabla^2 + \vec{v} + \vec{g} \quad (2.37)$$

in which the Coulomb exchange operator $g(x_1)$ is given by

$$\vec{g} = \hat{j} - \hat{k}. \quad (2.38)$$

Here,

$$J(x_1)f(x_1) = \sum_{k=1}^N \int \psi_k^*(x_2) \psi_k(x_2) \frac{1}{r_{12}} f(x_1) dx_2 \quad (2.39)$$

$$K(x_1)f(x_1) = \sum_{k=1}^N \int \psi_k^*(x_2) \psi_k(x_2) \frac{1}{r_{12}} f(x_1) dx_2 \quad (2.40)$$

with $f(x_1)$ an arbitrary function. The matrix ϵ consists of lagrange multipliers. Also,

$$\epsilon_{ji}^* = \epsilon_{ij}. \quad (2.41)$$

Where, ϵ is Hermitian. Now multiplying equation (2.30) with ψ_j^* and integrating, one obtains the formula for orbital energies

$$\epsilon = \epsilon_{ii} = \langle \psi_i | \hat{F} | \psi_i \rangle = H_i + \sum_{j=1}^N (J_{ij} - K_{ij}) \quad (2.42)$$

Summing over i and comparing with equation (2.33), we get

$$E_{HF} = \sum_{i=1}^N \epsilon_i - \hat{V}_{ee}. \quad (2.43)$$

Where the symbol $\hat{V}_{ee}$ stands for electron electron repulsion energy. Here,

$$\hat{V}_{ee} = \int \psi_{HF}^*(x^N) \left(\sum_{i < j} \frac{1}{r_{ij}} \right) \psi_H(x^N) dx^N \frac{1}{2} \sum_{i,j=1}^N (J_{ij} - K_{ij}). \quad (2.44)$$

For the total molecular energy including nucleus–nucleus repulsion one has,

$$W_{HF} = \sum_{i=1}^N \epsilon - \hat{V}_{ee} + \hat{V}_{nn}. \quad (2.45)$$

Neither E_{HF} nor W_{HF} is equal to the sum of orbital energies. Hartree–Fock method is a non-linear self-consistent field.

2.3.1 Limitations and Failings of the Hartree-Fock Approach

When the even number of electrons are located in double occupied spatial orbital, it is called that the compound is in singlet state. It also called closed-shell system. Again having odd number of electrons (compound with single occupied orbital) is called triplet state. It also called open shell system. This two types of system gives us two different approaches of Hartree-Fock method. In restricted HF (RHF) method, all electrons are considered to be paired where as in the unrestricted HF (UHF) method [53], this restriction is lifted totally.

Despite its advantages, the Hartree-Fock approximation has significant limitations. One of the primary drawbacks is that it assumes a mean-field approximation where each electron moves in an average field created by all other electrons. This simplification neglects the electron correlation effects, which are crucial in accurately describing the behavior of many-electron systems, particularly in cases where strong correlations are present. As a result, the Hartree-Fock method often leads to inaccuracies in calculated properties such as bond lengths, reaction energies, and excitation energies. Furthermore, while the Slater determinant effectively accounts for the antisymmetry of the wave function, it does not capture the dynamic correlation between electrons, which arises from their instantaneous interactions. Additionally, the computational cost of the Hartree-Fock method scales as $O(N^4)$, where N is the number of electrons in the system. This scaling arises because the method involves calculating integrals over all pairs of electron orbitals. As the number of electrons increases, the number of required calculations grows rapidly, making it computationally expensive for larger systems.

2.3.2 Correlation Energy

No single determinant or straightforward of a few determinants can ever accurately describe the wave function for a system with many interesting electrons. The calculation of the energy error, however, is here characterized as being negative, the difference between E_{HF} and E_{exact} is called correlation energy and can be expressed as [54],

$$E_{corr}^{HF} = E_{min} - E_{HF} \quad (2.46)$$

When atomic and molecular changes preserve the number and type of chemical bonds, correlation energy tends to remain constants, but it can fluctuate significantly and become decisive when bonds change. Its magnitude can range from a few hundredths of an atomic unit to hundreds of kilocalories per mole. Exchange energies are an order magnitude or more bigger, even if the self exchange term is omitted. Despite the fact that E_{corr} is usually small against E_{min} as in the example of N_2 molecule where

$$E_{corr}^{HF} = 14.9eV < 0.001E_{min} \quad (2.47)$$

It can have a huge influence. For instance, the experimental dissociation energy of N_2 molecule is

$$E_{diss} = 9.9eV < E_{corr} \quad (2.48)$$

which corresponds to a large contribution of the correlation energy to relative energies such as reaction energies which are of particular interest in quantum chemistry. The main contribution to the correlation energy arises from the mean field approximation used in the HF-method [55].

Chapter 3

Density Functional Theory

Density Functional Theory (DFT) is a powerful quantum mechanical modeling method widely used to investigate the electronic structure of many-body systems. Unlike traditional methods that rely on wave functions, DFT is based on the electron density, which is considered a fundamental variable. The key idea behind DFT is that all properties of a system can be determined from its electron density rather than the many-body wave function. This approach simplifies the problem of solving the Schrödinger equation, as it reduces the complexity associated with many-body interactions. DFT is particularly advantageous because it balances accuracy and computational efficiency, allowing researchers to study larger and more complex systems than would be feasible with wave-function-based methods. By leveraging the Hohenberg-Kohn theorems and the Kohn-Sham equations, DFT provides a practical framework for exploring a wide range of physical, chemical, and material properties.

3.1 The Electron Density

In DFT, the electron density $n(r)$ serves as the central variable for describing a quantum system. The electron density represents the probability of finding an electron at a particular point in space and can be derived from the many-body wave func-

tion $\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)$ by integrating over all electron coordinates. This relationship is expressed mathematically as [56]

$$n(\vec{r}) = N \int |\psi(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)|^2 d\vec{r}_2, \vec{r}_3, \dots d\vec{r}_N. \quad (3.1)$$

Additionally, we must remember that all electrons are identical; thus, we cannot label them as electron 1 or electron N . Instead, we can determine the probability of any order or set of N electrons being located at the coordinates $\vec{r}_1$ to $\vec{r}_N$. While the wave function contains comprehensive information about the quantum state of a system, it is the electron density that ultimately determines all measurable properties. The total number of electrons N in the system can also be calculated from the electron density using the equation [57]:

$$N = \int n(\vec{r}) d\vec{r}. \quad (3.2)$$

This integration highlights that the electron density encodes vital information about the total number of electrons, making it a fundamental aspect of DFT. By focusing on $n(\vec{r})$ instead of the complex multi-dimensional wave function, DFT simplifies calculations, making it a practical and efficient approach for studying the electronic structure of various materials.

3.2 Thomas-Fermi Model

Many people have looked into the subject of explaining the density of a assembly with multiple electrons, which led to the so-called density functional theory. The first exploration has done by Llewellyn Thomas and Enrico Fermi in 1927 [56, 58], which is known as Thomas-Fermi model. The model helps to describe the electronic structure of many election system. It was made in a semi-classical way soon after the Schrödinger equation was made. It's a semi-classical approach since it borrows some ideas from quantum mechanics. But the rest of the ideas don't use quantum physics. Instead, they can be operated with regular function. Unlike the wave-function-based approach, this formulation was completely based on electronic density and is seen as

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a precursor to the modern DFT. Let us consider the space divided into many small cubes, each of side l and volume $\Delta V = l^3$, each containing some fixed numbers of electrons ΔN and we assume that the electrons in each shell behave like independent fermions at the temperature 0 K, with the cells independent of one another. The energy level of a particle in a three dimensional infinite well are given by the formula.

$$\epsilon(n_x, n_y, n_z) = \frac{h^3}{8ml^2}(n_x^2 + n_y^2 + n_z^2) \quad (3.3)$$

$$= \frac{h^3}{8ml^2}R^2 \quad (3.4)$$

Where $(n_x, n_y, n_z) = 1, 2, 3 \dots$ and the second equality defines by the quantity R . For high quantum numbers, that is, for large R , the number of distinct energy levels with energy smaller than ϵ can be approximated by the volume of one octant of a spherical with radius R in the space n_x, n_y, n_z . This number is,

$$\phi(\epsilon) = \frac{1}{8} \left(\frac{4\pi R^3}{3} \right) \quad (3.5)$$

$$= \frac{\pi}{6} \left(\frac{8ml^2\epsilon}{h^2} \right)^{\frac{3}{2}} \quad (3.6)$$

The number of energy levels between ϵ and $\epsilon + \delta\epsilon$ is accordingly

$$g(\epsilon)\Delta\epsilon = \phi(\epsilon + \delta\epsilon) - \phi(\epsilon) \quad (3.7)$$

$$= \frac{\pi}{4} \left(\frac{8ml^2}{h^2} \right)^{\frac{3}{2}} \epsilon^{\frac{1}{2}}\delta\epsilon + O((\delta\epsilon)^2) \quad (3.8)$$

where the function $g(\epsilon)$ is the density of states at energy ϵ . To compute the total energy for the cell with electrons, we need the probability for the state with energy to be occupied which we call $f(\epsilon)$. This is the Fermi Dirac distribution.

$$f(\epsilon) = \frac{1}{1 + e^{\beta(\epsilon - \mu)}} \quad (3.9)$$

The Fermi energy ϵ_f is the zero temperature limit of the chemical potential μ . Now we find the total energy of the electrons in this cell by summing the contributions

from the different energy states.

$$\Delta E = 2 \int \epsilon f(\epsilon) g(\epsilon) d\epsilon \quad (3.10)$$

$$= 2 \int \epsilon f(\epsilon) \frac{\pi}{4} \left(\frac{8ml^2}{h^2} \right)^{\frac{3}{2}} \epsilon^{\frac{1}{3}} d\epsilon \quad (3.11)$$

$$= \frac{8\pi}{5} \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} l^3 \epsilon_f^{\frac{5}{2}} \quad (3.12)$$

where the factor 2 enters because each energy level is doubly occupied by one electron with spin and another with spin β . The fermi energy ϵ_f is related to the number of electrons ΔN in the cell through the formula:

$$\Delta N = 2 \int f(\epsilon) g(\epsilon) d\epsilon \quad (3.13)$$

$$= \frac{8\pi}{3} \left(\frac{2m}{h^2} \right)^{\frac{3}{2}} l^3 \epsilon_f^{\frac{3}{2}} \quad (3.14)$$

Eliminating ϵ_f from 3.12 and 3.14 we have,

$$\Delta E = \frac{3}{5} \Delta N \epsilon_f \quad (3.15)$$

$$= \frac{3h^2}{10m} \left(\frac{3}{8\pi} \right)^{\frac{2}{3}} l^3 \left(\frac{\Delta N}{l^3} \right)^{\frac{5}{3}} \quad (3.16)$$

Equation (3.14) is a relation between total kinetic energy and the electron density for each cell in the space. Adding the contribution from all cells we find the total kinetic energy to be, now reverting to atomic units,

$$T_T F[\rho] = C_F \int \rho^{\frac{5}{3}}(\vec{r}) d\vec{r} \quad (3.17)$$

where,

$$C_F[\rho] = \frac{3}{10} (3\pi^2)^{\frac{2}{3}} = 2.871 \quad (3.18)$$

Here, we first come across the LDA, one of the most significant concepts in contemporary density functional theory. By using locally applicable relations suited for a homogeneous electronic system, electronic characteristics are approximated as functions of the electron density. In terms of electron density, the energy formula

for an atom is

$$T_T F[\rho(\vec{r})] = C_F \int \rho^{\frac{5}{3}}(\vec{r}) d\vec{r} - Z \int \frac{\rho(\vec{r})}{r} + \frac{1}{2} \int \int \frac{\rho(\vec{r}_1)\rho(\vec{r}_2)}{r_1 - r_2} d\vec{r}_1 d\vec{r}_2 \quad (3.19)$$

This is the energy functional of Thomas-Fermi theory of atoms. The method became considered as an overly simplified model of little real significance for quantitative predictions in atomic, molecular, or solid state physics because the accuracy for atoms is not as high with this model as it is with other methods.

3.3 Hohenberg-Kohn (HK) Theorems

Density functional theory as we know it today was born in 1964 when a landmark paper by Hohenberg and Kohn appeared in the Physical Review [59]. The theorems they introduced represent the major theoretical pillars on which all modern-day density functional theories are built. These theorems laid the groundwork for the fundamental relationship between electron density and the properties of quantum systems, making DFT a powerful alternative to wave-function-based methods.

The Hohenberg-Kohn theorems are central to the formulation of DFT, and they can be summarized as follows:

3.3.1 Theorem 1

Statement: The ground-state electron density $n(\vec{r})$ uniquely determines the external potential $V_{ext}(\vec{r})$ acting on the electrons. This means that if the electron density of a system is known, the external potential can be uniquely inferred, allowing for the derivation of all ground-state properties, including the total energy, from the electron density.

According to the first theorem, the ground-state density and the external potential correspond in a one-to-one manner. Since the external potential is fixed, the Hamiltonian, and hence the wave function ψ , is determined by the ground-state density $n_0(\vec{r})$.

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Proof: Consider the ground states of two N -electron systems, characterized by two different external potentials $V_{ext}(\vec{r})$ and $V'_{ext}(\vec{r})$, which differ by more than an additive constant. The corresponding Hamiltonians, $\hat{H}$ and $\hat{H}'$, would both have the same ground-state density $n(\vec{r})$, but different ground-state wave functions [60], ψ and ψ' , with $\hat{H}\psi = E_0\psi$ and $\hat{H}'\psi' = E'_0\psi'$. Since ψ' is not the ground state of $\hat{H}$, it follows that

$$E_o < \langle \psi' | \hat{H} | \psi' \rangle \quad (3.20)$$

$$< \langle \psi' | \hat{H}' | \psi' \rangle + \langle \psi' | \hat{H} - \hat{H}' | \psi' \rangle \quad (3.21)$$

$$< E'_o + \int n_o(\vec{r})[V_{ext}(\vec{r}) - V'_{ext}(\vec{r})]d\vec{r} \quad (3.22)$$

Similarly,

$$E'_o < \langle \psi | \hat{H}' | \psi \rangle \quad (3.23)$$

$$< \langle \psi | \hat{H} | \psi \rangle + \langle \psi | \hat{H}' - \hat{H} | \psi \rangle \quad (3.24)$$

$$< E_o + \int n_o(\vec{r})[V'_{ext}(\vec{r}) - V_{ext}(\vec{r})]d\vec{r} \quad (3.25)$$

By adding equation 3.6 and 3.9 we get,

$$E_o + E'_o < E'_o + E_o \quad (3.26)$$

Hence, n_o two different external potentials $V_{ext}(\vec{r})$ can give rise to the same ground state density $n_0(\vec{r})$ which determines the external potential $V_{ext}(\vec{r})$, except for a constant. That is to say, there is a one-to-one mapping between the ground state density $n_0(\vec{r})$ and the external potential $V_{ext}(\vec{r})$, although the exact formula is unknown.

3.3.2 Theorem 2

Statement: For any trial electron density $n(\vec{r})$, the energy functional $E[n]$ will yield a value that is greater than or equal to the ground-state energy E_0 . The equality holds when the trial density corresponds to the true ground-state density. This variational principle implies that one can minimize the energy functional $E[n]$

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with respect to the electron density to find the ground state of a system.

Proof: There exists a universal functional $F[n(\vec{r})]$ of the density, independent of the external potential $V_{ext}(\vec{r})$, such that the minimum value of the energy functional

$$E[n(\vec{r})] = \int n(\vec{r})V_{ext}(\vec{r})d\vec{r} + F[n(\vec{r})] \quad (3.27)$$

yields the exact ground-state energy of the system. The exact ground-state density $n_0(\vec{r})$ minimizes this functional. Thus, the exact ground-state energy and density are fully determined by the functional $E[n(\vec{r})]$. The universal functional $F[n(\vec{r})]$ can be written as:

$$F[n(\vec{r})] \equiv \hat{T}[n(\vec{r})] + E_{int}[n(\vec{r})] \quad (3.28)$$

where,

$\hat{T}[n(\vec{r})]$ = kinetic energy

$E_{int}[n(\vec{r})]$ = interaction energy of the particles.

According to variational principle, the energy functional $E[\psi']$ is

$$E[\psi'] \equiv \langle \psi' | \hat{T} + \hat{V}_{int} + \hat{V}_{ext} | \psi' \rangle \quad (3.29)$$

reaches its global minimum only when ψ' is the ground-state wave function ψ_o , with the constraint that the total number of particles is conserved. According to the first theorem of HF, ψ' must correspond to a ground state with particle density $n'(\vec{r})$ and external potential V'_{ext} , making $E[\psi']$ a functional of $n'(\vec{r})$. According to the variational principle,

$$E[\psi'] \equiv \int n'(\vec{r})V'_{ext}d\vec{r} + F[n'(\vec{r})] > E[\psi_o] = \int n(\vec{r})V_{ext}d\vec{r} + F[n_o(\vec{r})] = E[n_o(\vec{r})] \quad (3.30)$$

Hence, energy functional $E[n(\vec{r})] \equiv \int n(\vec{r})V_{ext}d\vec{r} + F[n_o(\vec{r})]$ evaluates for the correct ground state density n_o is lower than the value of the function of any other density $n(\vec{r})$. By minimizing the total energy functional with respect to the variations in the density $n(\vec{r})$, one could find the exact ground state density and energy. This

functional, however, only determines ground-state properties and does not provide any insight into excited states.

3.3.3 Advantage and Disadvantage of HK Theorems

The Hohenberg-Kohn theorems are fundamental to density functional theory and offer significant advantages in computational chemistry and materials science. One of the primary benefits is their ability to relate the ground-state properties of many-electron systems directly to electron density, simplifying calculations compared to wave-function-based methods. The first theorem establishes a unique mapping between the ground-state electron density and the external potential, ensuring that all ground-state properties can be derived from the electron density alone. Additionally, the second theorem introduces a variational principle that allows for the efficient optimization of electron density to find the ground state. By shifting the focus from finding a function of $3N$ variables (the wave function) to a function of three variables (the electron density), the Hohenberg-Kohn theorems significantly simplify the process of resolving the Schrödinger equation. These theorems make DFT a versatile tool applicable to a wide range of systems, from small molecules to large biological complexes.

However, the Hohenberg-Kohn theorems also come with notable disadvantages. They are limited to ground-state properties, providing no direct insight into excited states, which poses challenges in studying electronic excitations and charge transfer processes. The effectiveness of DFT is highly dependent on the choice of exchange-correlation functional, which may not accurately capture all correlation effects, especially in systems with strong electron-electron interactions. Furthermore, while DFT is generally computationally less intensive than wave-function methods, the calculations of exchange-correlation energies can still be demanding. The reliance on external potentials and the interpretation challenges associated with electron density further highlight the limitations of the Hohenberg-Kohn theorems, necessitating careful consideration when applying DFT to complex systems.

3.4 Kohn-Sham Equation

Hohenberg and Kohn framework is not very usefull yet in actual calculation. The second Kohenberg-Kohn theorem is the only possibility for the minimization of energy. An example of an iterative approachis the Hartree equation for self consistent single particle [41,60]. The Hartree equation is clearly a wavefunction based equation and it is not directly related to Hohenberg and Kohn. Hartree's approximation says that every electron moves in an effective single particle potential of the form of,

$$v_H(\vec{r}) = -\frac{Z}{|\vec{r}|} + \int \frac{n(\vec{r}')}{|\vec{r} - \vec{r}'|} d\vec{r}'. \quad (3.31)$$

where, $-\frac{Z}{|\vec{r}|}$ = attractive Coulomb potential of a nucleus with an atomic number Z. $\int \frac{n(\vec{r}')}{|\vec{r} - \vec{r}'|} d\vec{r}'$ = correspond to the potential caused by the mean electron density distribution $n(\vec{r})$. $n(\vec{r})$ can be expressed in terms of the single particle wave function,

$$n(\vec{r}) = \sum_{j=1}^M |\phi_j|^2. \quad (3.32)$$

According to the Pauli exclusion principle, the sum of above equation runs over the lowest eigenvalues. For single particle. $3N$ -dimensional Schrödinger equation can be written as,

$$[-\frac{1}{2}\vec{\nabla}^2 + v_H(\vec{r})] \phi_j(\vec{r}) = \epsilon_j \phi_j(\vec{r}) \quad j = 1, \dots, N. \quad (3.33)$$

Therefore Khon and Sham investegated the DFT applied to a system of N non-interacting electron in an external potential. The expression for the energy of such a system is,

$$E_{v(\vec{r})}[n'(\vec{r})] \equiv \int v(\vec{r})n'(\vec{r})d\vec{r} + T_s[n'(\vec{r})] \geq E. \quad (3.34)$$

where, $n'(\vec{r}) = v$ respectable density for non-interacting electrons $T_s[n'(\vec{r})]$ = kinetic energy of the ground state of those non-interacting electron. In case of non-interacting electron, the Euler-Lagrange equation can be written as,

$$\delta E_v[n'(\vec{r})] \equiv \int \delta n'(\vec{r})[v(\vec{r}) + \frac{\delta}{\delta n'(\vec{r})}T_s[n'(\vec{r})]|_{n'(\vec{r})=n(\vec{r})} - \epsilon]d\vec{r} = 0. \quad (3.35)$$

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where, $n'(\vec{r})$ = exact ground state density for the potential $v(\vec{r})$. ϵ = the Lagrangian multiplier to ensure particle density conservation. For such a system the total ground state energy and particle density can simply be written as,

$$E = \sum_{j=1}^N \epsilon_j. \quad (3.36)$$

and

$$n(\vec{r}) = \sum_{j=1}^N |\phi_j(\vec{r})|^2. \quad (3.37)$$

For an interacting case the construction of Euler-Lagrange equation becomes.

$$\delta E_v[n'(\vec{r})] \equiv \int \delta n'(\vec{r}) [v_{eff}(\vec{r}) + \frac{\delta}{\delta n'(\vec{r})} T_s[n'(\vec{r})]|_{n'(\vec{r})=n(\vec{r})} - \epsilon] d\vec{r} = 0. \quad (3.38)$$

with

$$v_{eff}(\vec{r}) \equiv v(\vec{r}) + \int \frac{n(\vec{r}')}{|\vec{r} - \vec{r}'|} d\vec{r}' + v_{xc}(\vec{r}). \quad (3.39)$$

and the functional derivative

$$v_{xc}(\vec{r}) \equiv \frac{\delta}{\delta n'(\vec{r})} E_{xc}[n'(\vec{r})]|_{n'(\vec{r})=n(\vec{r})}. \quad (3.40)$$

The corresponding equations are the single particle Schrödinger equation

$$[-\frac{1}{2}\vec{\nabla}^2 + v_{eff}(\vec{r})] \phi_j(\vec{r}) = \epsilon_j \phi_j(\vec{r}) \quad j = 1, \dots, N. \quad (3.41)$$

As well as the equation for the particle density is

$$n(\vec{r}) = \sum_{j=1}^N |\phi_j(\vec{r})|^2. \quad (3.42)$$

Which form together self consistent Kohn-Sham equation and effective potential. If one use the exact $E_{xc}[n(\vec{r})]$ and $v_{xc}[n(\vec{r})]$ it would be possible to find the exact solution.

3.5 Solving Kohn-Sham (KS) Equation

Solving the Kohn-Sham equation is a crucial step in density functional theory for obtaining the ground state electron density in a many body system. In a condensed matter system, the Kohn-Sham equation gives a way to obtain the exact density and energy of the ground state. The process starts with an initial electron density $n(\vec{r})$, usually a superposition of atomic electron density, then the effective KS potential V_{eff} is calculated and the KS equation is solved with single-particle eigenvalues and wave functions, a new electron density is then calculated from the wave functions. This iterative process is commonly performed numerically through self-consistent

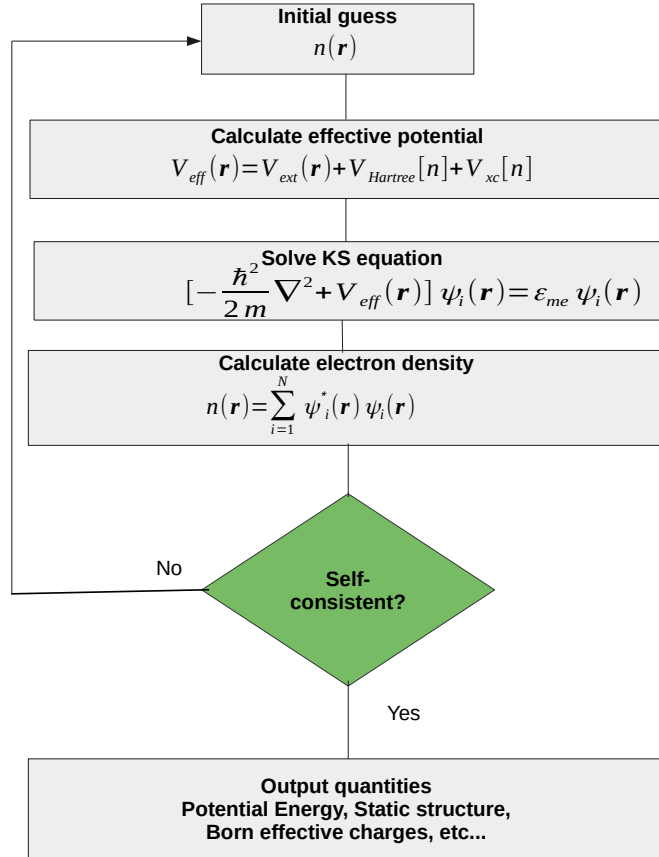


Figure 3.1: Flowchart illustrating the iterative process for solving the Kohn-Sham equation in DFT to obtain the ground-state electron density and associated properties of a many body system.

iterations, as illustrated in the flowchart Figure 3.1. The self-consistency can be determined by monitoring the change in total energy, electron density from the

previous iteration, or the total force acting on the atoms being less than a specified small quantity, or by using a combination of these criteria. If self-consistency is not achieved, the calculated electron density is mixed with the density from previous iterations to derive a new electron density, and the process repeats. This continues until self-consistency is reached. Once convergence is achieved, various quantities can be calculated, including total energy, forces, stress, eigenvalues, electron density of states, and band structure, among others.

3.6 The Exchange-Correlation (XC) Functional

In density functional theory, the exchange-correlation potential, $V_{xc}[n(r)]$, is a critical component that captures the complex quantum mechanical effects of electron exchange and correlation. The exchange–correlation is a word for how the electrons in a material interact with each other. It combines the effects of exchange and correlation, which are two basic ideas in quantum physics that explains how electrons interact with each other. The exchange potential comes from the fact that electrons are identical objects and follow the Pauli exclusion principle, which says that it is important for two fermions that are identical to inhibit the same quantum state simultaneously. The correlation potential come from the fact that electron connect with each other through Coulombic forces, which depend on where and how fast they are moving. In DFT, to solve the Kohn–Sham hamiltonian, we also need an expression for the exchange-correlation potential. For the solution different theoretical models and estimates are used to get close to the exchange correlation potential. The accuracy of this rough estimates varies on the type of material being modeled and how it is used. In the next parts, we will talk about and analyze the local density and generalizes gradient approximations, which are two of the most common ways to solve the exchange-correlation functional.

3.6.1 Local Density Approximation (LDA)

The local density approximation (LDA) is one of the most common approximation, here V_{xc} is considered to depend solely on the local electron density. This approx-

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imation is based on the idea that in regions where the electron density is nearly uniform, the behavior of electrons can be likened to that of a homogeneous electron gas. While simple, LDA can perform reasonably well for certain materials, especially solids with nearly uniform electron densities. Walter Kohn described LDA as the forebearer of all approaches in his nobel lecture. From this approximation, the exchange-correlation energy follows the formula [61]:

$$E_{xc}^{LDA}(n(\vec{r})) = \int n(\vec{r})\epsilon_{xc}(n(\vec{r}))d\vec{r} \quad (3.43)$$

From spin polarized systems, the form of the equation is change into:

$$E_{xc}^{LSDA}(n_{\uparrow}(\vec{r}), n_{\downarrow}(\vec{r})) = \int n(\vec{r})\epsilon_{xc}(n_{\uparrow}(\vec{r}), n_{\downarrow}(\vec{r}))d\vec{r} \quad (3.44)$$

Here, LSDA means local spin density approximation. The exchange-correlation energy $\epsilon_{xc}(n(\vec{r}))$ per particle at each point in the system is the same as that of uniform electron gas of same density. The exchange-correlation energy can be divided into two parts, the exchange part and the correlation part. Using the equation, we find the exchange-correlation potential as:

$$V_{xc}^{LDA} = \frac{\delta E_{xc}^{LDA}(n(\vec{r}))}{\delta n} = \epsilon_{xc}(n(\vec{r})) + n(\vec{r})\frac{\delta \epsilon_{xc}(n(\vec{r}))}{\delta n} \quad (3.45)$$

and the Khon-Sham equations read,

$$[-\frac{1}{2}\nabla^2 + \nabla(\vec{r}) + \int \frac{n(\vec{r}')}{|\vec{r} - \vec{r}'|}d\vec{r}' + V_{xc}^{LDA}(\vec{r})]\psi = \epsilon_i\psi_i. \quad (3.46)$$

This self consistent solution defines the KS local density approximation, which is the literature is usually simply called local density approximation (LDA) method. The function $\epsilon_{xc}(n)$ can be devided into exchange and correlation contributions,

$$\epsilon_{xc}(n) = \epsilon_x(n) + \epsilon_c(n). \quad (3.47)$$

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The exchange part is already known given by the Dirac exchange energy functional.

$$\epsilon_x(n) = -C_x n^{\frac{1}{3}}(\vec{r}). \quad (3.48)$$

Where,

$$C_x = \frac{3}{4} \left(\frac{3}{\pi} \right)^{\frac{1}{3}}. \quad (3.49)$$

The advantage of LDA is its low complexity, which makes it computationally effective and makes it possible to examine big systems. LDA is the only case where the exchange-correlation functional can be derived exactly. It has been discovered that this approximation is fairly true for a variety of systems, especially for bulk solids. An additional advantage of this approximation is that LDA sheds light on the electrical characteristics of materials, enabling researchers to forecast and comprehend phenomena including chemical reactivity, optical characteristics, and magnetic behavior. This approximation has many disadvantages as well. The solutions of exchange correlation effects in the LDA are improper when the assumption of an electron gas with a slowly varying density as the starting point is fundamentally false. Examples include the Van der Waals interactions between non-overlapping subsystems, the electronic Wigner crystal, and electronic tails dissipating into space on the surfaces of bounded electronic systems. However, density functional theory may effectively address such issues when using the right approximations.

3.6.2 Generalized Gradient Approximation (GGA)

The LDA neglects the inhomogeneties of the real charge density which could be very different from the homogenous electron gas (HEG). The exchange correlation (xc) energy of inhomogeneous charge density can be significantly different from the HEG result. This leads to be the development of various generalized-gradient approximations (GGA) [62] which include density gradient corrections and higher spatial derivatives of the electron density and give better result than LDA in many cases. Three most widely used GGA's are the from proposed by Becke, Perdew et al., Burke and Enzerhof. The definition of the xc energy functional of GGA is the

generalized form in the equation of LSDA to include corrections [63]

$$E_{xc}^{LDA}[n_{\downarrow}(\vec{r}), n_{\uparrow}(\vec{r})] = \int n(\vec{r}) \epsilon_{xc}^{home}[n_{\downarrow}(\vec{r}), n_{\uparrow}(\vec{r})]. \quad (3.50)$$

Where, xc energy density $\epsilon_{xc}^{home}[n(\vec{r})]$ is a function of the density alone and is composed into exchange energy density $\epsilon_{xc}^{home}[n(\vec{r})]$ and correlation energy density ϵ_C^{home} . So that the xc energy functional is decomposed into exchange energy function $E_{xc}^{LDA}[n(\vec{r})]$ linearly. From density gradient $\nabla(\vec{r})$ as,

$$\begin{aligned} E_{xc}^{GGA}[n_{\downarrow}(\vec{r}), n_{\uparrow}(\vec{r})] &= \int n(\vec{r}) \epsilon_{xc}^{home}[n_{\downarrow}(\vec{r}), n_{\uparrow}(\vec{r}), |\nabla_{\uparrow}(\vec{r})|, |\nabla_{\downarrow}(\vec{r})|, \dots] d\vec{r}. \quad (3.51) \\ &= \int n(\vec{r}) \epsilon_{xc}^{home} n(\vec{r}) F_{xc}[n_{\downarrow}(\vec{r}), n_{\uparrow}(\vec{r}), |\nabla_{\uparrow}(\vec{r})|, |\nabla_{\downarrow}(\vec{r})|, \dots] d\vec{r}. \end{aligned} \quad (3.52)$$

Where, F_{xc} is dimensionless and $\epsilon_{xc}^{home} n(\vec{r})$ is the exchange energy density of the unpolarized HEG. F_{xc} can be decomposed linearly into exchange contribution $F_{xc} = F_x + F_c$. Generally, GGA works better than LDA, in predicting binding energy of molecules and bond length, crystal lattice constants, especially the system where charge density varied rapidly. In case of ionic crystal, GGA overcorrects LDA results where the lattice constants of LDA fit well than GGA. But in case of transition metal oxides and rare-earth element, both LDA and GGA perform badly. This drawback leads to approximations beyond LDA and GGA.

GGA is better than LDA at predicting the electronic qualities of materials because it takes both the local electron density and the gradient of that density into account. It also gives a better explanation of how molecules and solids are held together by chemical bonds. It can also predict non-local features like magnetic interactions and exchange-correlation energies. GGA also needs less computer power than more advanced ways, which makes it easier to use for large-scale calculations. Overall, GGA has made it much easier to simulate and understand how materials behave at the atomic level.

3.6.3 Local Spin Density Approximation (LSDA)

In the majority of the exchange-correlation functionals, the homogenous electron gas is used. Particularly, in local density approximation, the density, n is considered at each point in space and the homogenous electron gas model is applied locally using the density only [64]. In the local spin density approximation (LSDA), the electron density for individual spin components, $n_\alpha(\vec{r})$ and $n_\beta(\vec{r})$ the homogeneous electron gas model, the exchange energy functional is known as Dirac exchange and has a very simple mathematical form

$$E_x^{LSDA} = -\frac{3}{2} \left(\frac{3}{4\pi} \right)^{\frac{1}{3}} \int [n_\alpha(\vec{r})^{\frac{4}{3}} + n_\beta(\vec{r})^{\frac{4}{3}}] d\vec{r}. \quad (3.53)$$

The general form of the local spin density approximation for the correlation energy functional has the following form

$$E_c^{LSDA}[n_\alpha + n_\beta] = \int n[\vec{r} \epsilon_c(n_\alpha(\vec{r}), n_\beta(\vec{r}))] d\vec{r}. \quad (3.54)$$

where, $\epsilon_c[n_\alpha(\vec{r}), n_\beta(\vec{r})]$ is the correlation energy per electron in a homogeneous electron gas. The exact form is unknown but many approximations to $\epsilon_c[n_\alpha(\vec{r}), n_\beta(\vec{r})]$ exist. LSDA gives fairly good results for equilibrium geometries and vibrational frequencies for covalently bonded molecules but has a tendency to overbind atoms because the molecule is overstabilized compared to the separate atoms. Moreover, the results are bad for molecules containing hydrogen bonds and Van der Waals complexes.

In LSDA, the exchange energy is typically underestimated by 10% and the correlation energy overestimated by 100%. The total energy is too high, the gap between occupied and unoccupied orbitals (or bands in solid state physics) is too low. Moreover, LSDA favors the $d^{n+1}s^1$ configuration over the $d^n s^2$ in the 3d transition metal atoms. Gunnarsson and Jones showed that the major source of error in LSDA comes from the exchange energy. A source of error in the calculation of the exchange energy is the self-interaction repulsion present in LSDA. This error is related to the fact that an electron sees all other electrons including itself. The presence of such an error is

easy to verify, since the exchange energy of any one-electron system should be zero. Perdew and Zunger have proposed a self-interaction corrected LSDA functional that reduced the error below 3% for the exchange energy. Improvement was also obtained in the LSDA correlation energy, in the total energy, in the orbital eigenvalues, in the long range behavior of $V_{xc}(\vec{r})$, in the shape of the exchange-correlation hole and consequently in the electronic density.

3.6.4 LDA+U Method

Strongly correlated system usually contain transition metal or rare-earth metal ions with partially filled d or f shells. Because of the orbital-independent potentials in LSDA and GGA, they cannot properly describe such systems. The total energy in LSDA+U method [65] is given by,

$$E_{tot}^{LDA}[n_\sigma(\vec{r}), n_\sigma] = E^{LDSA}[n_\sigma(\vec{r})] + E^U[n_\sigma] - E_{dc}[n(\vec{r})]. \quad (3.55)$$

where, σ = spin indexes, $n(\vec{r})$ = electron density for spin- σ electrons, n_σ = density matrix of d or f electron for spin- σ electrons, $E^{LDSA}[n_\sigma(\vec{r})]$ = standard LSDA energy functional, and $E^U[n(r)]$ = electron-electron coulomb interaction energy. The last term is double counting term which remove the average LDA energy contribution of d or f electrons from the LDA energy.

$$E_{dc}[n(\vec{r})] = \frac{1}{2}UN(N-1) - \frac{1}{2}J[N_\uparrow(N_\uparrow-1) + N_\downarrow(N_\downarrow-1)]. \quad (3.56)$$

Where, $N = N_\uparrow + N_\downarrow$. U and J are coulomb and exchange parameters. If exchange and non-sphericity is neglected then,

$$E_{tot}^{LDA+U} = E_{LDA} + \frac{1}{2}U \sum_{i \neq j} n_i n_j - \frac{1}{2}UN(N-1) \quad (3.57)$$

The orbital energies ϵ_i are derivative of above equation with respect to orbit occupations n_i

$$\epsilon_i = \frac{\partial E}{\partial n_i} = \epsilon_{LDA} + U\left(\frac{1}{2} - n_i\right) \quad (3.58)$$

For $n_i = 1$, LDA orbital energies are shifted by $-\frac{U}{2}$ and by $\frac{U}{2}$ and by for unoccupied orbitals ($n_i=0$), resulting the upper and lower Hubbard bands, which opens a gap at the Fermi energy in transition metal oxides. In case of double counting term, it has two different treatment: around mean-field (AMF) and fully localized limit (FLL). The former is most suitable for small U system and the latter for large U system. The energies for double counting is given by,

$$E_{dc}^{AMF} = \frac{1}{2}UN^2 - \frac{U + 2lJ}{2l - 1} \frac{1}{2} \sum_{\sigma} N_{\sigma}^2 \quad (3.59)$$

and

$$E_{dc}^{AMF} = \frac{1}{2}UN(N - 1) - \frac{1}{2}J \sum_{\sigma} N_{\sigma}(N_{\sigma} - 1) \quad (3.60)$$

3.6.5 Hybrid Functional

Hybrid functional approximation is a sort of computational method employed in DFT studies. It was created by A. Becke [66]. Hybrid functionals try to get around this problem by mixing parts of both the local density approximation (LDA), and the generalized gradient approximation (GGA) [67]. In this method, the first exchange-correlation energy was written as,

$$E_{xc}^{hyb} = \gamma E_{xc}^{KS} + (1 - \gamma) E_{xc}^{GGA} \quad (3.61)$$

Here, E_{xc}^{KS} is the exchange energy calculated with the exact Kohn-Sham wavefunctions and γ is known as fitting parameter. There are numerous popular hybrid functionals available, including B3LYP, PBE0, and HSE06. The main benefit of hybrid functionals is that they blend the best parts of both LDA and GGA functionals. LDA does a good job of describing how the density changes slowly, but it doesn't account for the exchange-correlation energy in systems where the density changes. GGA, on the other hand, gives a better picture of the exchange-correlation energy, but it often overestimates or underestimates certain properties, such as band-gaps, bond lengths, and reaction energies. Hybrid functionals get around these problems by adding a small amount of exact Hartree-Fock exchange to the normal GGA func-

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tional. This makes the predicted properties more accurate, especially for systems with big bandgaps, states that are localized, and atoms of transition metals.

In general, the benefit of hybrid functional approach is better than LDA and GGA functionals because it is more accurate and stable. This quality makes it handy for predicting the properties of materials and molecules in a wide range of applications, such as catalysis, energy storage, and electronic devices.

Results and Discussion

4.1 Computational Details

In this study, first principles calculation were performed using the full potential linearized augmented plane wave (FP-LAPW) method based on density functional theory (DFT) [68] as implemented in WIEN2k code [69]. The structural optimization was performed to compute the optimum ground state of the studied material, utilizing the Perdew–Burke–Ernzerhof (PBE) with the generalized gradient approximation (GGA) [70] method. For non-overlapping sphere, the muffin-tin radii of the cubic $\text{Ba}_2\text{NbRhO}_6$ double perovskite was taken to 2.50, 1.93, 2.02, and 1.74 a.u. (atomic units) for Ba, Nb, Rh, and O, respectively. The self-consistent calculations converged with respect to the total energy convergence of 10^{-5} Ry and the total charge convergence of $10^{-3}e$. The cut-off parameters $R_{MT} \times K_{max} = 8$ was used, where R_{MT} is the smallest muffin–tin radius and K_{max} is the largest reciprocal lattice vector used in the expansion of the plane wave functions. The Gaussian factor was set to $G_{max} = 12 \text{ (a.u.)}^{-1}$, while the angular momentum vector, $l_{max} = 10$. An energy cut-off -6 Ry has been selected to separate the core and valence states. Self-consistent calculations were performed using $10 \times 10 \times 10$ k-mesh in the first Brillouin zone to analyze structural properties. A denser mesh of $49 \times 49 \times 49$ was selected to

enhance the accuracy of the computed optical properties. Elastic properties have been investigated using Charpin's approach [71]. Thermoelectric properties were evaluated using the BoltzTraP code [72], which is based on the constant relaxation time approximation. In addition, the famous Slack's model was used to estimate the lattice thermal conductivity of the material, offering crucial insights into their thermal properties.

4.2 Structural Properties

The studied compound $\text{Ba}_2\text{NbRhO}_6$ has a cubic crystalline structure with a space group $Fm\bar{3}m$ (#225) is illustrated in Figure 4.1. In this structure, the Ba, Nb, Rh, and O atoms are located at 8c(0.25, 0.25, 0.25), 4a(0, 0, 0), 4b(0.5, 0, 0), and 24e(0.5, 0.2539, 0) Wyckoff positions.

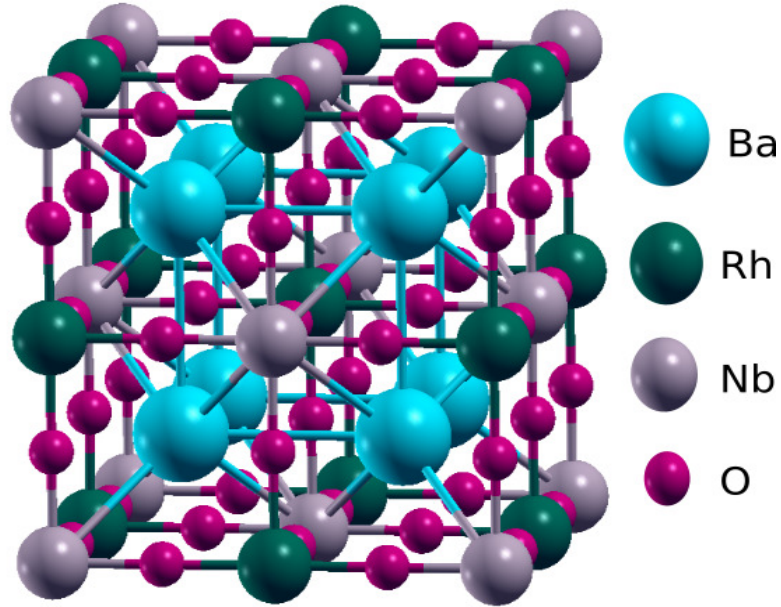


Figure 4.1: Conventional unit cell of $\text{Ba}_2\text{NbRhO}_6$ double perovskite.

Volume optimization performed to obtain the lattice parameters of the ground state configuration with minimum energy value of the system. The variation of total energy with respect to unit cell volume is depicted in Figure 4.2 and the optimized lattice parameters are obtained by fitting the data to the Murnaghan equation of

Results and Discussion

state, which can be written as [73],

$$E(V) = E_o + \frac{B_o V}{B'_o} \left[\frac{\left(\frac{V_o}{V}\right)^{B'_o}}{B'_o - 1} + 1 \right] - \frac{B_o V_o}{B'_o - 1}. \quad (4.1)$$

Where, V_o and V denote the ground state and deformed unit cell volumes, while E_o , B_o and B'_o are the ground state energy, bulk modulus and its derivative. The value of the lattice constant of $\text{Ba}_2\text{NbRhO}_6$ double perovskite under different pressures has been shown in Table 4.1. The values show that as the pressure increases, both the lattice parameter and the cell volume decrease gradually, accompanied by a reduction in bond lengths. Due to the impact of pressure, the distances among the constituent atoms Ba–Nb, Rh–O, Nb–O, and Ba–O decrease.

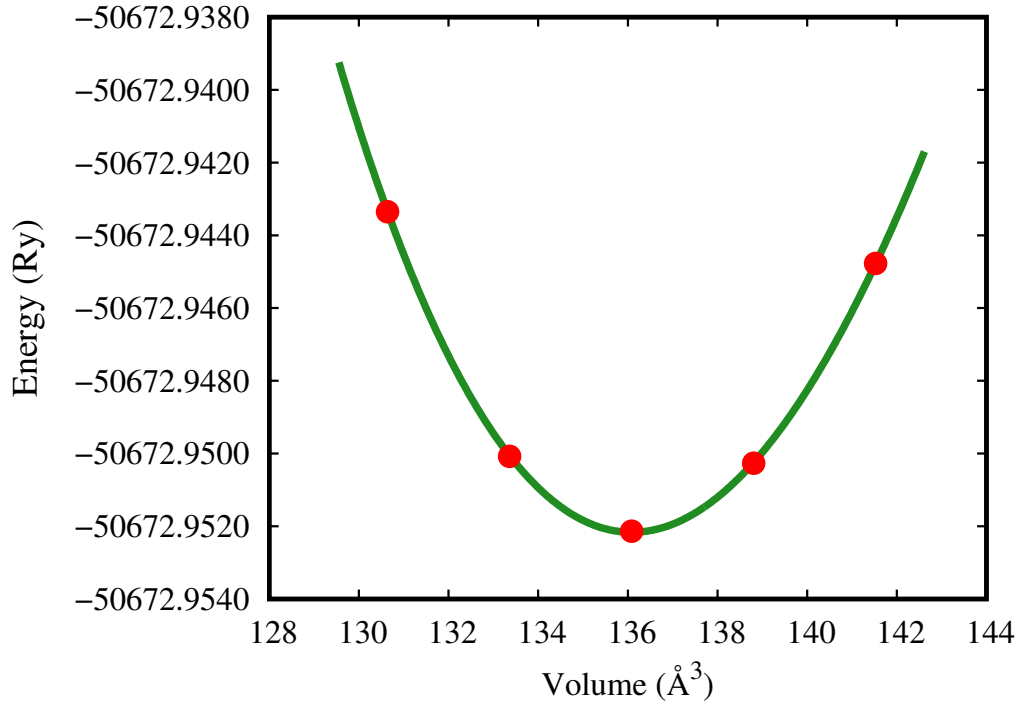


Figure 4.2: Energy versus volume optimization curve of $\text{Ba}_2\text{NbRhO}_6$ double perovskite.

Goldsmith's tolerance factor (t_F) and octahedral factor (μ) have been calculated to assess the structural stability of the studied material, which are estimated by the equations

$$t_F = \frac{R_{Ba} + R_O}{\sqrt{2} \left(\frac{R_{Nb} + R_{Rh}}{2} + R_O \right)}. \quad (4.2)$$

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Table 4.1: Computed lattice parameter a_o , volume a_o^3 , and bond length of $\text{Ba}_2\text{NbRhO}_6$ double perovskite under different hydrostatic pressures.

Pressures (GPa)	a_o (Å)	a_o^3 (Å ³)	Bond length (Å)			
			Ba-Nb	Rh-O	Nb-O	Ba-O
0	8.165	544.338	3.536	2.073	2.010	2.887
	8.165 [20]	—	3.53 [20]	2.07 [20]	2.01 [20]	2.88 [20]
5	8.087	528.886	3.502	2.053	1.991	2.860
10	8.021	515.849	3.473	2.036	1.974	2.836
20	7.911	495.101	3.426	2.009	1.947	2.797
40	7.749	465.304	3.356	1.957	1.907	2.740
60	7.631	444.369	3.304	1.937	1.878	2.698
80	7.538	428.320	3.264	1.914	1.855	2.665
100	7.462	415.495	3.231	1.894	1.837	2.638

and

$$\mu = \frac{R_{Nb} + R_{Rh}}{2R_O}. \quad (4.3)$$

Here, R_{Ba} , R_{Nb} , R_{Rh} , and R_O represents the ionic radii of cation Ba , Nb , Rh and that of anion O , respectively [74]. The cubic double perovskite is considered perfectly stable when t_F lies between 0.89 and 1.00 [75]. The estimated tolerance and octahedral factors of $\text{Ba}_2\text{NbRhO}_6$ are 0.91 and 0.53, respectively. To compute thermodynamic stability of the material, formation energy (ΔH) has been calculated using the formula

$$\Delta H = \frac{E_{\text{Ba}_2\text{NbRhO}_6} - 2E_{Ba} - E_{Nb} - E_{Rh} - 6E_O}{10}. \quad (4.4)$$

Where, ΔH is the formation energy, $E_{\text{Ba}_2\text{NbRhO}_6}$ is the total energy of the systems,

E_{Ba} , E_{Nb} , E_{Rh} and E_O are the total energies of Ba, Nb, Rh, and O atoms, respectively. The calculated formation energy of the compound is -3.48 eV/atom, confirming that the compound is thermodynamically stable because negative value of a system indicates its stability. The good agreement of our computed structural parameters with previously investigated experimental results of oxide double perovskite materials [20] confirms the reliability of our presented results.

4.3 Electronic Properties

The calculation of electronic properties of the studied material along with band structure and density of states provide valuable information for understanding its optical properties. To fully comprehend the photoelectric conversion efficiency of perovskite materials, it is essential to analyze their electronic structure. This includes calculating the energy band structure, total density of states, and partial density of states of the investigated double perovskite materials under varying hydrostatic pressures, ranging from 0 to 100 GPa.

4.3.1 Band Structure

The electronic band structure is a representation of the allowed electronic energy levels of materials and is used to better inform their electrical properties. The electronic band structure is crucial for identifying band types of a material. The plot of the computed band structure related to the wave vector and the energy function within the first Brillouin zone are illustrated in Figure 4.3 and 4.4. The black horizontal dashed line at 0 eV represents the Fermi level (E_F) and the valence band (VB) and conduction band (CB) are presented by colored line below and above the E_F , respectively. Here, the calculation is done by defining highly symmetric points on the edge of the Brillouin zone with sampling path of W-L- Γ -X-W-K. The band structure of the studied material shows that the conduction band minimum (CBM) is located at the Γ point, while the valence band maximum (VBM) is found at the X symmetry point. Since CBM and VBM lie at different symmetry points, it is understood that the studied compound Ba_2NbRhO_6 has an indirect bandgap

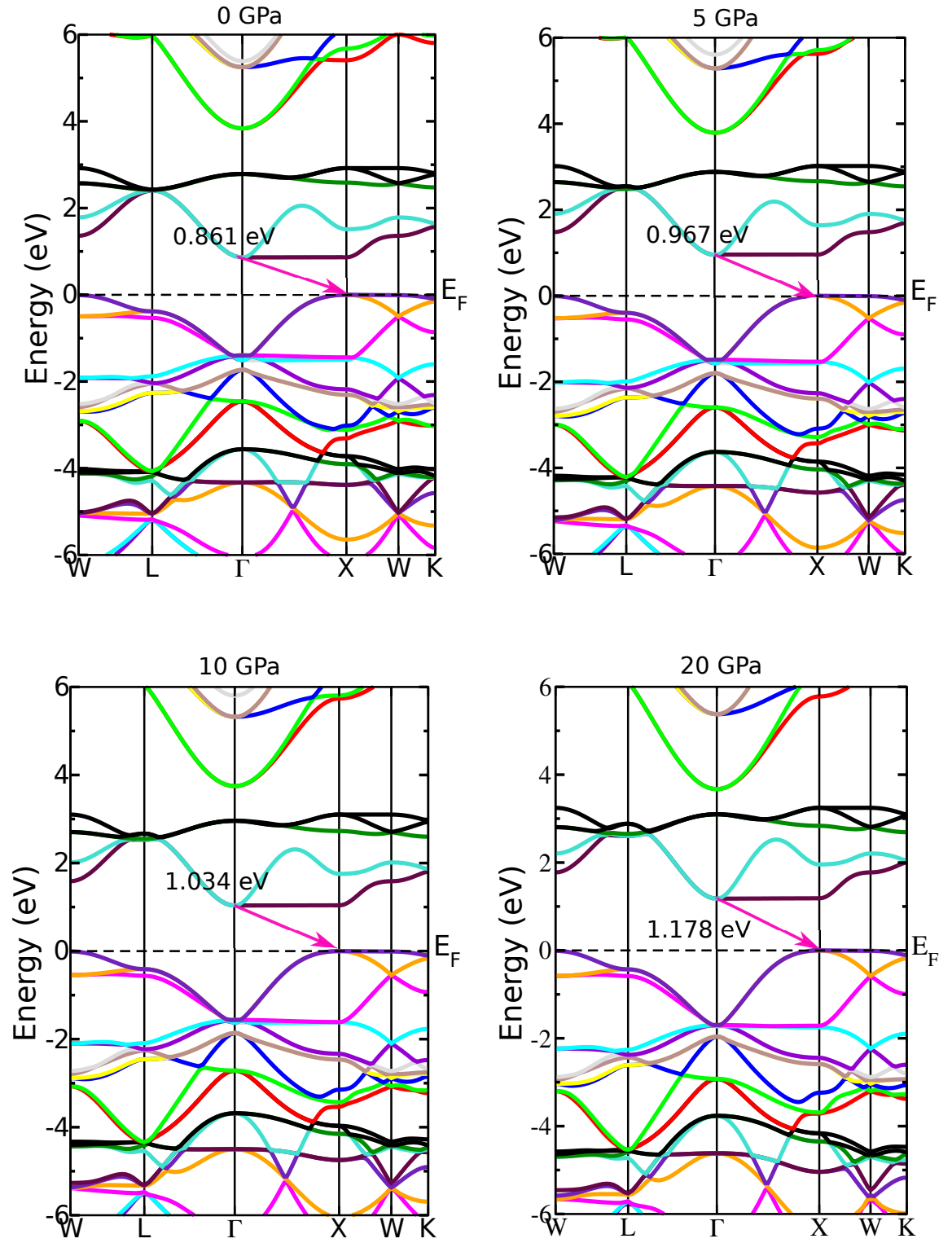


Figure 4.3: Band structure of $\text{Ba}_2\text{NbRhO}_6$ double perovskite at ambient and different hydrostatic pressures.

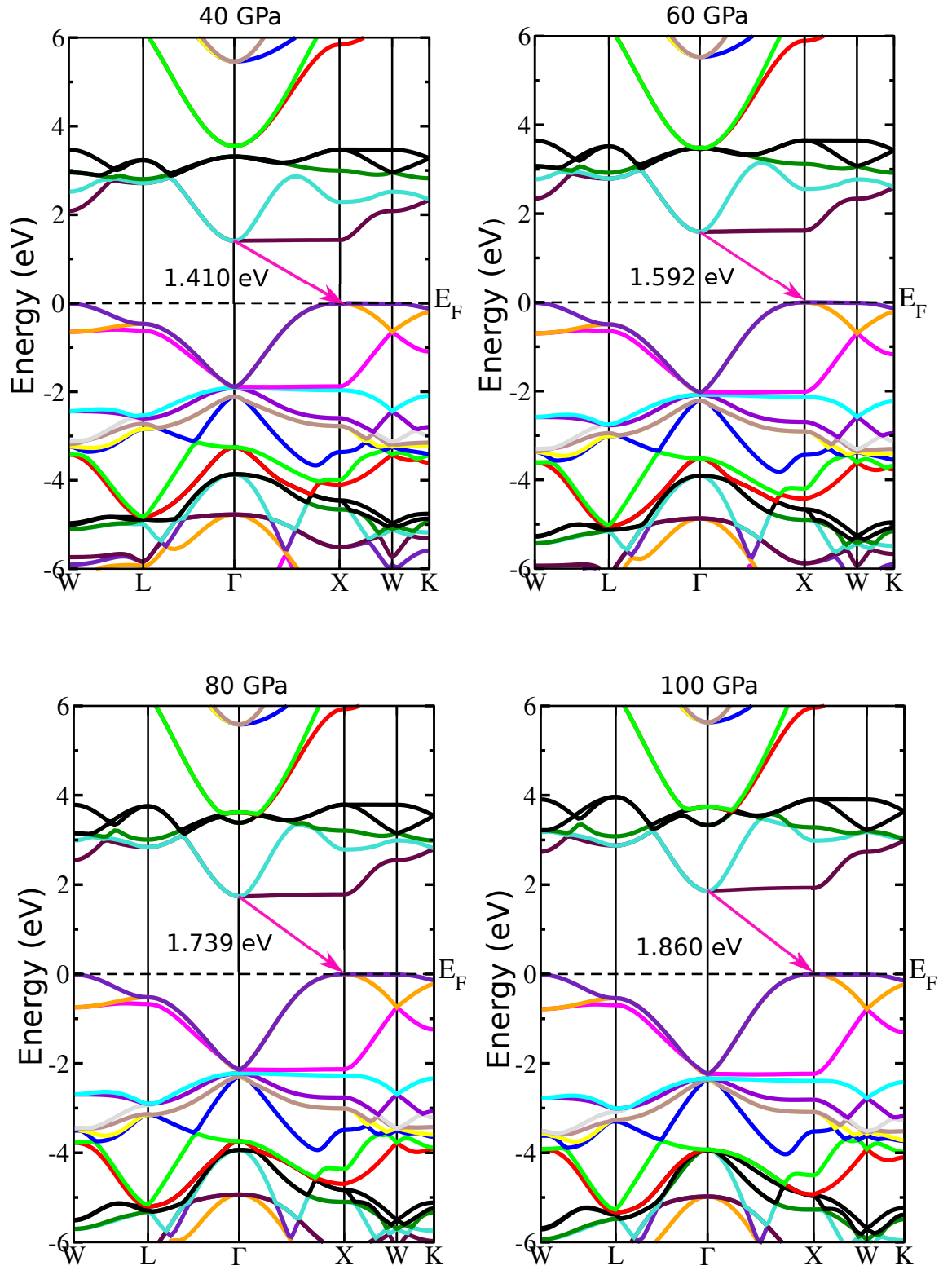


Figure 4.4: Band structure of $\text{Ba}_2\text{NbRhO}_6$ double perovskite at ambient and different hydrostatic pressures.

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of 0.861 eV at ambient pressure, which is presented in Figure 4.3. The results reveal an inverse relation between the lattice constant and the band gap. when pressure is applied, the lattice constant decreases because the pressure compresses the lattice dimensions and enhances orbital interactions. As a result, the existing states between the VBM and CBM exert a repulsive interaction, which increases the band gap. Finally, The bandgap at 100 GPa is found to be 1.860 eV, indicating a larger forbidden gap between the VB and CB compared to other applied pressures.

Table 4.2: Computed bandgap E_g (eV), effective mass of the hole m_h^* (m_o), effective mass of the electron m_e^* (m_o), and relative effective mass ratio D of $\text{Ba}_2\text{NbRhO}_6$ double perovskite under different hydrostatic pressures.

Pressures (GPa)	E_g (eV)	m_h^* (m_o)	m_e^* (m_o)	D
0	0.861	0.377	0.237	1.591
	0.874 [20]	—	—	—
5	0.967	0.367	0.235	1.562
10	1.034	0.362	0.234	1.547
20	1.178	0.347	0.231	1.502
40	1.410	0.339	0.226	1.498
60	1.592	0.312	0.220	1.418
80	1.739	0.299	0.214	1.397
100	1.860	0.288	0.210	1.371

Effective mass quantifies the dynamic response of charge carriers (electrons or holes) in a crystal lattice under the influence of external forces. The effective mass of electrons m_e^* (m_o) and holes m_h^* (m_o) within the conduction and valence bands were

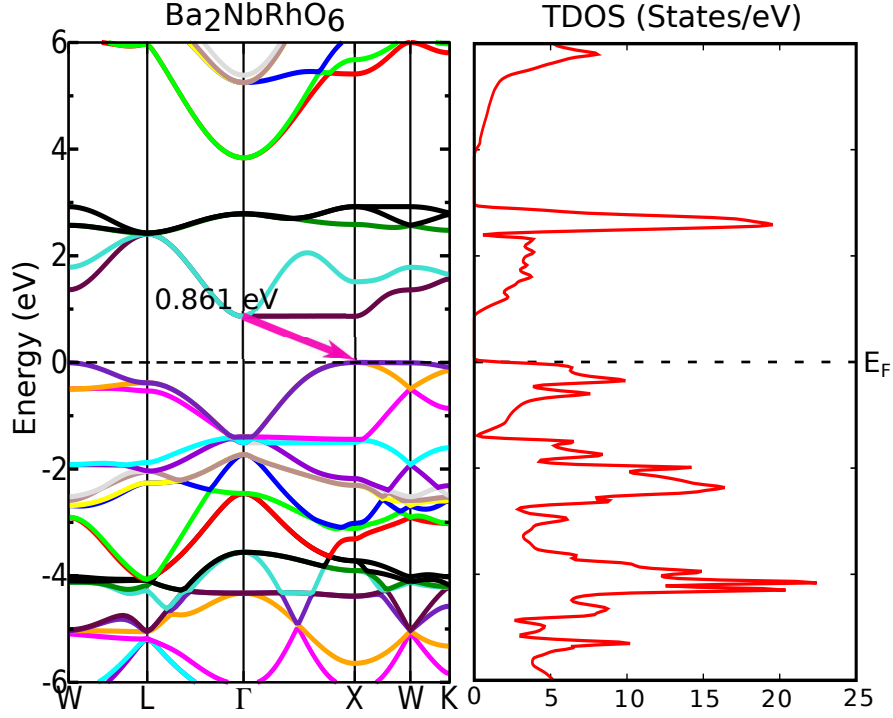


Figure 4.5: Band structure and total density of states at ambient pressure.

estimated using the formula

$$m^*(k) = \hbar^2 \times \left(\frac{d^2\epsilon}{dk^2} \right)^{-1}. \quad (4.5)$$

Here, m^* , $\hbar$, and ϵ represent the effective mass of the carrier, the reduced Planck constant, and the energy eigenvalue of the material [76]. The effective mass of charge carriers increases as the electronic band structure becomes flatter or less curved, owing to the inverse relationship between effective mass and band curvature. As presented in Table 4.2, $\text{Ba}_2\text{NbRhO}_6$ exhibits a significantly large carrier effective mass at ambient pressure, which corresponds to the nearly flat nature of the VBM, indicating the presence of heavy holes. In contrast, at 100 GPa, the band structure becomes more dispersive near the VBM, resulting in a notably lower carrier effective mass. In addition, the relative effective mass ratio (D) was calculated using the expression, $D = m_h^*(m_o)/m_e^*(m_o)$. Where, m_o denotes the rest mass of a free electron [77]. Table 4.2 reveals that the computed values of D consistently exceed unity across the pressure range, indicating the material's potential for use in

photocatalysts.

4.3.2 Density of States

To obtain deep insight visualization of electronic states of the studied material, the total density of states (TDOS) and partial density of states (PDOS) have been plotted as a function of energy. The TDOS and PDOS are commonly utilized to emphasize the contributions of different states to the formation band, which also characterize the transport properties of a material. TDOS also describes the contribution of electronic states toward VB and CB, and accurately interprets the size of bandgap, which is illustrated in Figure 4.5. The total and partial DOS exhibit the individual

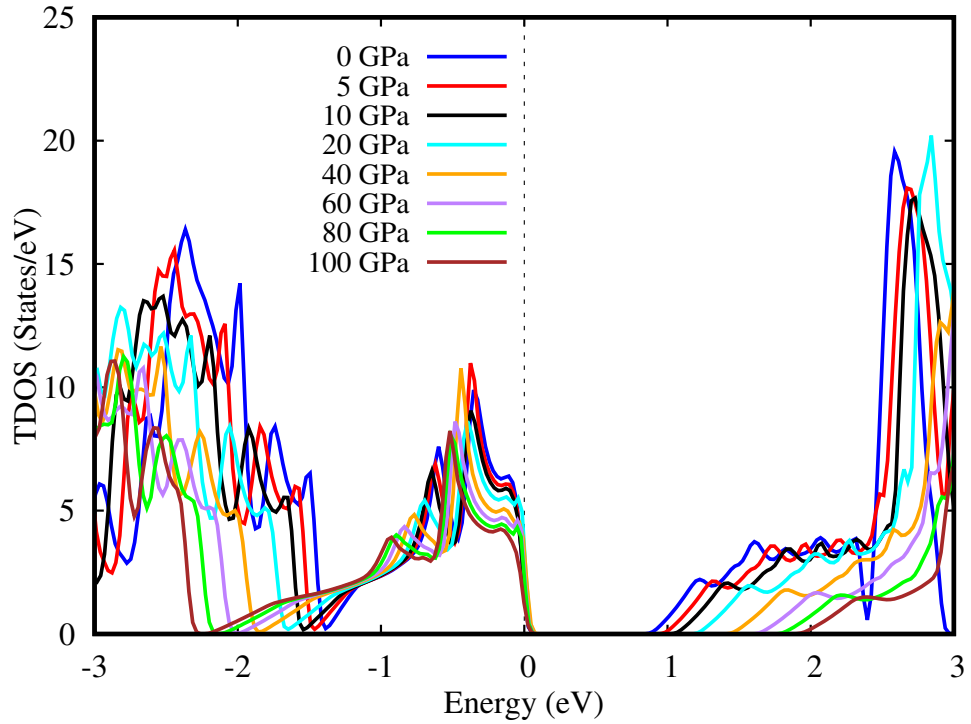


Figure 4.6: Total density of states of $\text{Ba}_2\text{NbRhO}_6$ double perovskite under hydrostatic pressures.

contributions of different atoms in the VB and CB, are illustrated in Figure 4.6 and 4.7 for various hydrostatic pressures. At ambient and pressurized condition, Ba-3p, Nb-3d, Rh-3d and O-2p make up majority contribution to formation of the VB between the energy range of -3 to 0 eV, while Rh-3d mainly contribute to the VB between 0.9 and 3 eV. The bandgap increases continuously with pressure from

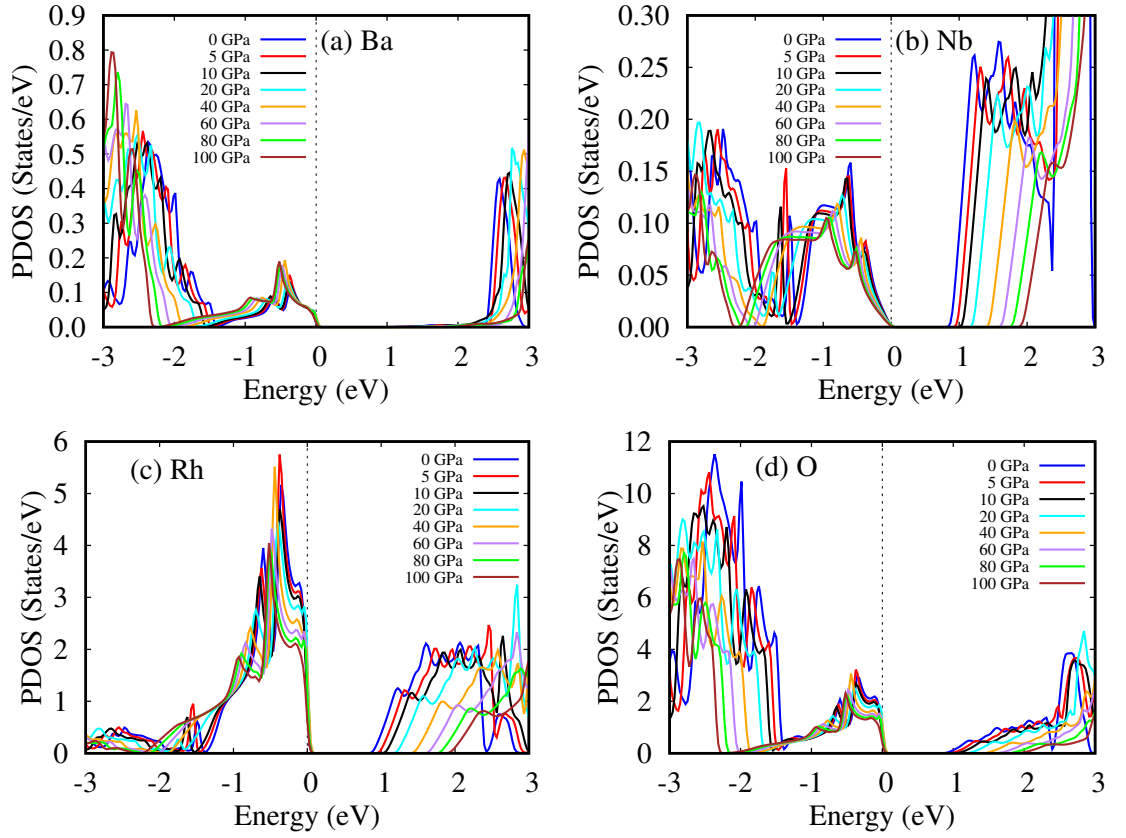


Figure 4.7: Partial density of states (a) Ba, (b) Nb, (c) Rh, and (d) O of Ba₂NbRhO₆ double perovskite under various hydrostatic pressures.

0 to 100 GPa, due to strong hybridization between Ba-3d and Nb-3d orbitals in the CB. The figure also demonstrated that TDOS varies at the E_F with pressure. This variation can be correlated with the bandgap, which also changes with pressures. TDOS plot illustrates that the contribution decreases with induced pressure both in VB and CB and it also display the contribution in the CB shifts to the right from the E_F , which proved that the studied material is a p-type semiconductor. In contrast, the PDOS shows that Ba's contribution in the VB increases with pressure, while its conduction in the CB shifts away from the E_F , but Nb, Rh, and O represent opposite trend, the contribution in CB shifts toward the E_F , as a result, the separation between the VB and CB respect to the E_F is also influenced. Notably, Nb and Rh have consistently made the most significant contributions to the VB and CB under all applied pressure.

4.4 Optical Properties

The study of optical properties is vital for exploring potential applications in photovoltaic and other optoelectronic devices. The optical properties offer valuable insights into how a material interacts with light, specifically its capacity to absorb, reflect, and transmit electromagnetic (EM) radiation. We carried out detailed calculations of the optical properties of $\text{Ba}_2\text{NbRhO}_6$ double perovskite under varying hydrostatic pressures, including the dielectric function, refractive index, extinction coefficient, absorption coefficient, optical conductivity, reflectivity, and energy loss function.

4.4.1 Dielectric Functions

The optical nature of any material is explained by the dielectric function $\epsilon(\omega)$, consisting of two parts which are the real dielectric function $\epsilon_1(\omega)$ and the imaginary dielectric function $\epsilon_2(\omega)$, and is given by Ehrenreich and Cohen's equation as follows [78]:

$$\epsilon(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega). \quad (4.6)$$

Here, ω symbolizes the angular frequency of EM radiation incident on the specimen. The real dielectric function $\epsilon_1(\omega)$ can be determined using the Kramers-Kronig equation [77],

$$\epsilon_1(\omega) = 1 + \frac{2}{\pi}P \int_0^\infty \frac{\omega' \epsilon_2(\omega')}{(\omega')^2 - (\omega)^2} d\omega'. \quad (4.7)$$

Here, P represents the principal quantum number. The real part of $\epsilon(\omega)$ represents the material's polarization response when interacting with an electromagnetic wave. The value of $\epsilon_1(\omega)$ can be either positive or negative: negative values indicate metallic behavior, while positive values exhibit semiconductor or insulator behavior. Figure 4.8(a) demonstrates that the value of $\epsilon_1(\omega)$ increases from its static value with increasing photon energy and exhibits two prominent peaks in visible and ultraviolet (UV) region, at ambient and different hydrostatic pressures. The largest peak occurs in the energy ranges from 2.79 eV to 3.47 eV for all applied pressures.

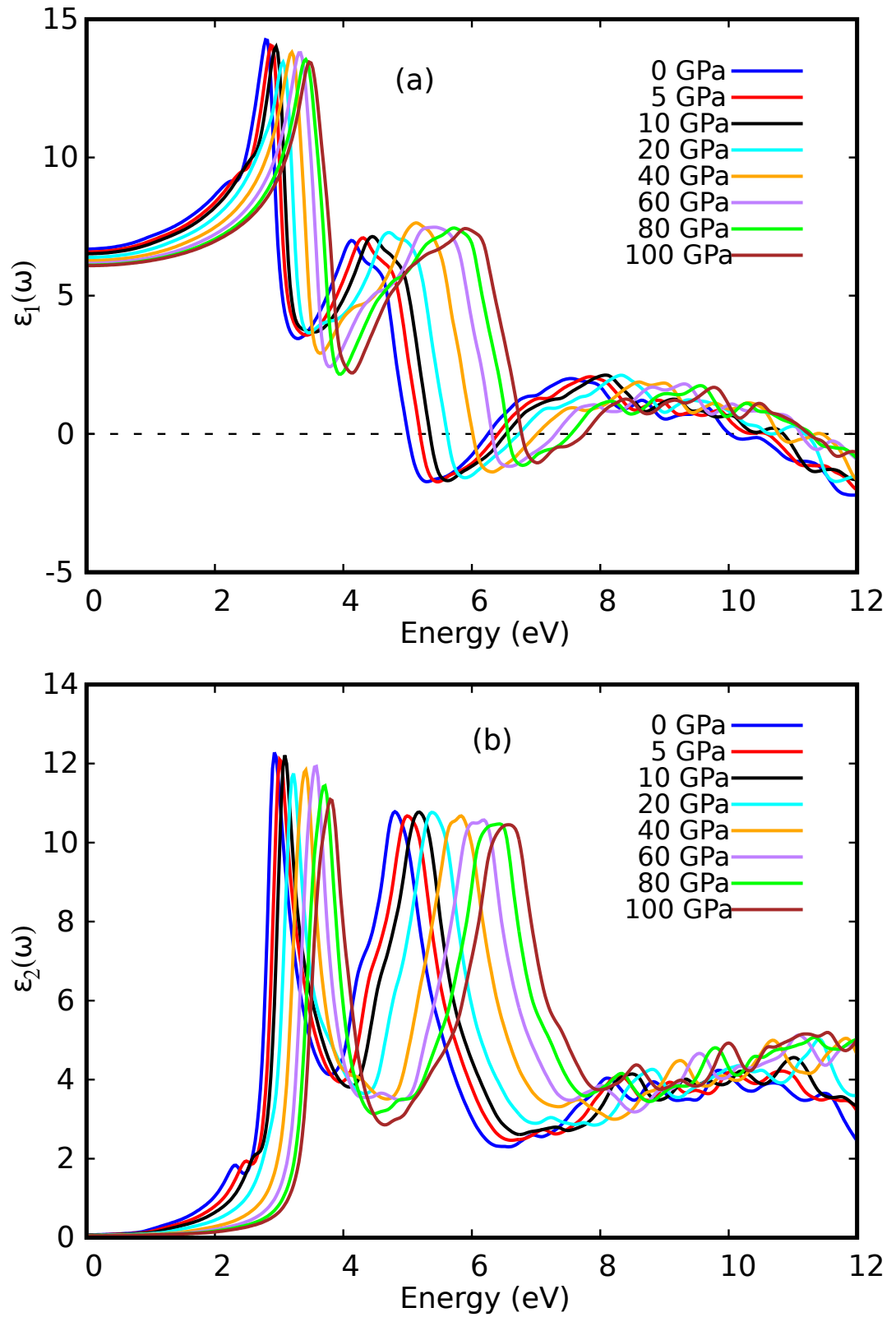


Figure 4.8: Optical properties of Ba₂NbRhO₆ double perovskite including (a) Real dielectric function $\epsilon_1(\omega)$ and (b) Imaginary dielectric function $\epsilon_2(\omega)$ under different pressures.

Results and Discussion

The real part of the dielectric function exhibits negative values in the energy ranges of 5.02 eV to 6.78 eV and above 10.03 eV at all pressure conditions. The presence of negative values of $\epsilon_1(\omega)$ confirms the metallic behavior of the studied material. The imaginary dielectric function $\epsilon_2(\omega)$ can be approximated as [79]:

$$\epsilon_2(\omega) = \frac{e^2 h}{\pi \omega^2 m^2} \sum_{v,c} \int |M_{cv}(k)|^2 \delta[\omega_{cv}(k) - \omega] d^3k. \quad (4.8)$$

Where, h denotes Planck's constant, k is the wave vector, ω is the angular frequency, and M is the molar mass of the carriers. The imaginary part of $\epsilon(\omega)$ reflects the material's light absorption characteristics. Figure 4.8(b) depicts $\epsilon_2(\omega)$ as a function of photon energy, showing two pronounced peak in visible and UV region for all exerted pressures, indicating maximum absorption. The threshold energy increases with rising pressure, progressing from 0.872 eV to 1.875 eV at 100 GPa which closely aligns with the bandgap energy.

4.4.2 Refractive Index

The refractive index $n(\omega)$ plays a vital role in device fabrication. The refractive index describes how much light bends and slows down when passing through a material. Higher refractive indices indicate more significant bending and slower light speeds. The relationship between the refractive index $n(\omega)$ as a function of frequency and the dielectric constant can be expressed by the equation,

$$n(\omega) = \frac{1}{\sqrt{2}} [(\epsilon_1^2(\omega) + \epsilon_2^2(\omega))^{\frac{1}{2}} + \epsilon_1(\omega)]^{\frac{1}{2}} \quad (4.9)$$

Figure 4.9(a) illustrates the variation of $n(\omega)$ and photon energy under hydrostatic pressures upto 100 GPa. When photons enter a material, it slows down due to the interaction with electrons, therefore the value of $n(\omega)$ is greater than unity [80]. The higher the refractive index of a material, the more the photon slows down as it passes through the material. The static refractive index, which characterizes the material's response at low frequencies and exhibits an inverse relationship with the energy bandgap [81].

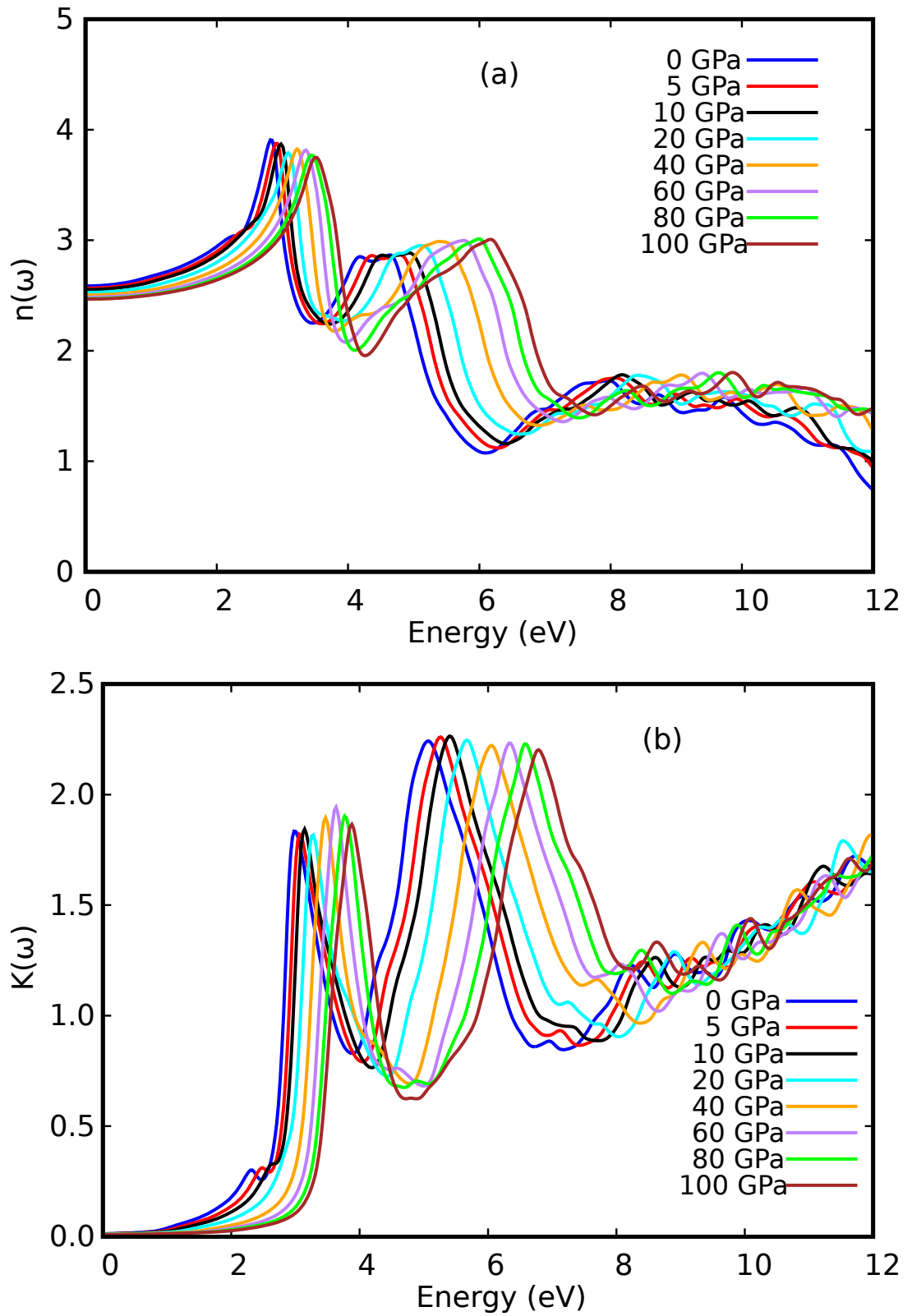


Figure 4.9: The pressure induced (a) Refractive index $n(\omega)$ and (b) Extinction coefficient $K(\omega)$ of $\text{Ba}_2\text{NbRhO}_6$ double perovskite, at ambient and pressurized conditions.

Results and Discussion

The computed value of $n(\omega)$ are 2.59, 2.57, 2.55, 2.53, 2.50, 2.48, 2.47, and 2.47 for 0, 5, 10, 20, 40, 80, and 100 GPa, respectively, shown in Table 4.3. At various applied pressures, the value of $n(\omega)$ increases linearly in the energy range of 0–3.52 eV and attains a prominent peak in the visible region after which it shifts to the UV region. Figure 4.8(a) and 4.9(a) reveal a notable similarity between the spectra of $\epsilon_1(\omega)$ and $n(\omega)$. Since $n(\omega)$ has a value less than one in the energy range of 12 to 13.56 eV under varying pressure. As a result, the group velocity shifts into the negative region, and the properties of the medium change from linear to nonlinear [82].

4.4.3 Extinction Coefficient

The extinction coefficient is an optical properties which explains the hindering in the amplitude of the electric vector's oscillations of incident radiation and it also offers useful insights into the level of absorption that an electromagnetic wave encounters as it travels through the material. The relationship between the extinction coefficient and the absorption coefficient can be mathematically expressed using the following formula [20]

$$K(\omega) = \frac{\alpha\lambda}{4\pi}. \quad (4.10)$$

Where, λ is the wavelength of light in nanometers or centimeters. Figure 4.9(b) is depicted the extinction coefficient $K(\omega)$ with the energy of incident radiation. The plot of $K(\omega)$ demonstrates two peaks: one in the visible region and another in the UV region, with the most prominent peak consistently occurring in the UV region across all applied pressures.

4.4.4 Absorption Coefficient

The absorption coefficient $\alpha(\omega)$ is the amount of incident radiation a material can absorb. It describes the efficiency of solar power conversion and indicates how far a specific energy can penetrate a material before being absorbed. Materials with highest $\alpha(\omega)$ value play a crucial role in optoelectronic and photovoltaic devices. The absorption coefficient can be represented by the dielectric constant using the

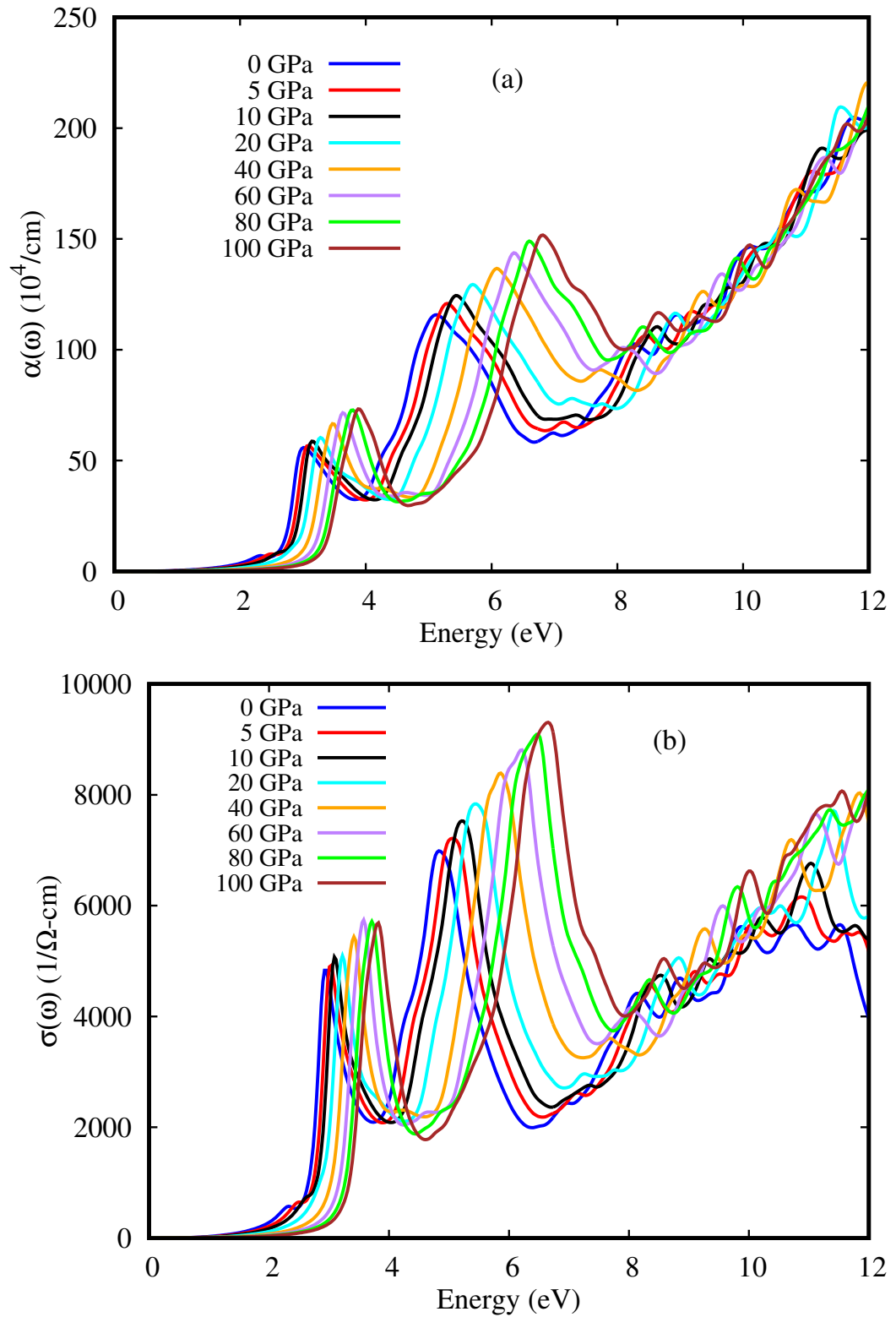


Figure 4.10: Optical properties of $\text{Ba}_2\text{NbRhO}_6$ double perovskite including (a) Absorption coefficient $\alpha(\omega)$ and (b) Optical conductivity $\sigma(\omega)$ under different hydrostatic pressures.

equation [83]

$$\alpha(\omega) = \sqrt{2}\omega \left(\sqrt{\epsilon_1^2(\omega) + \epsilon_2^2(\omega)} - \epsilon_1(\omega) \right)^{\frac{1}{2}}. \quad (4.11)$$

The evolution of the absorption spectra for the studied material is depicted in Figure 4.9(a). Absorption mainly begins at the threshold energy, which is also known as the absorption edge. The threshold energy of $\text{Ba}_2\text{NbRhO}_6$ rises with induced pressures and this value has consistent with the band gap value. The largest absorption peaks range from 5.13 eV to 6.82 eV at all hydrostatic pressure values considered. Figure 4.10(a) exhibits that the maximum value of $\alpha(\omega)$ was observed at a hydrostatic pressure of 8 GPa, while the lowest value occurred at 0 GPa, indicating its suitability for ultraviolet optoelectronic applications under pressure.

4.4.5 Optical Conductivity

Optical conductivity is another essential parameter that characterizes a material's electromagnetic response. The optical conductivity can be computed by the following equation [84],

$$\sigma(\omega) = \frac{\omega}{4\pi} \epsilon_2(\omega) \quad (4.12)$$

The optical conductivity $\sigma(\omega)$ spectra for the incident radiation energy is shown in Figure 4.10(b), the value of $\sigma(\omega)$ increases with pressure representing two peaks in the visible and UV region, with the highest peaks observed at 4.83 eV and 6.65 eV in the UV range under pressures of 0 GPa and 100 GPa, respectively. It has been observed that the optical conductivity and absorption display their minima and maxima in the same region of electromagnetic field, which confirms the theoretical concept and the accuracy of the computed results.

4.4.6 Optical Reflectivity

Optical reflectivity describes how much light is reflected off the surface of a material and can be used to analyze its surface properties. For studying the reflected light

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Table 4.3: Calculated static value of real dielectric function $\epsilon_1(0)$, reflectivity $R(0)$, refractive index $n(0)$, and maximum conductivity for $\text{Ba}_2\text{NbRhO}_6$ at ambient and pressurized conditions.

Pressures (GPa)	Real dielectric function $\epsilon_1(0)$	Reflectivity $R(0)$	Refractive index $n(0)$	Maximum conductivity $\sigma(\omega)(\Omega.\text{cm})^{-1}$
0	6.71	0.19	2.59	6987.82
5	6.58	0.19	2.57	7212.29
10	6.52	0.19	2.55	7529.00
20	6.38	0.19	2.53	7835.58
40	6.27	0.18	2.50	8390.88
60	6.17	0.18	2.48	8810.31
80	6.10	0.18	2.47	9091.56
100	6.09	0.18	2.47	9307.11

from the surface, reflectivity $R(\omega)$ can be calculated by the equation [84],

$$R(\omega) = \left[\frac{\sqrt{\epsilon_1(\omega) - i\epsilon_2(\omega)} - 1}{\sqrt{\epsilon_1(\omega) + i\epsilon_2(\omega)} + 1} \right]^{\frac{1}{2}}. \quad (4.13)$$

The graphical representation of $R(\omega)$ as a function of photon energy is illustrated in Figure 4.11(a). With increasing photon energy, $R(\omega)$ slightly rises and shows an almost similar nature in the UV region under all applied pressures. The lower $R(\omega)$ in the visible spectrum enhances the potential of metal halides for optoelectronic device applications.

4.4.7 Energy Loss Function

Energy loss function $L(\omega)$ characterizes the energy decay of a moving electron within a material [85].

$$L(\omega) = \frac{\epsilon_2(\omega)}{\epsilon_1(\omega)^2 + \epsilon_2(\omega)^2}. \quad (4.14)$$

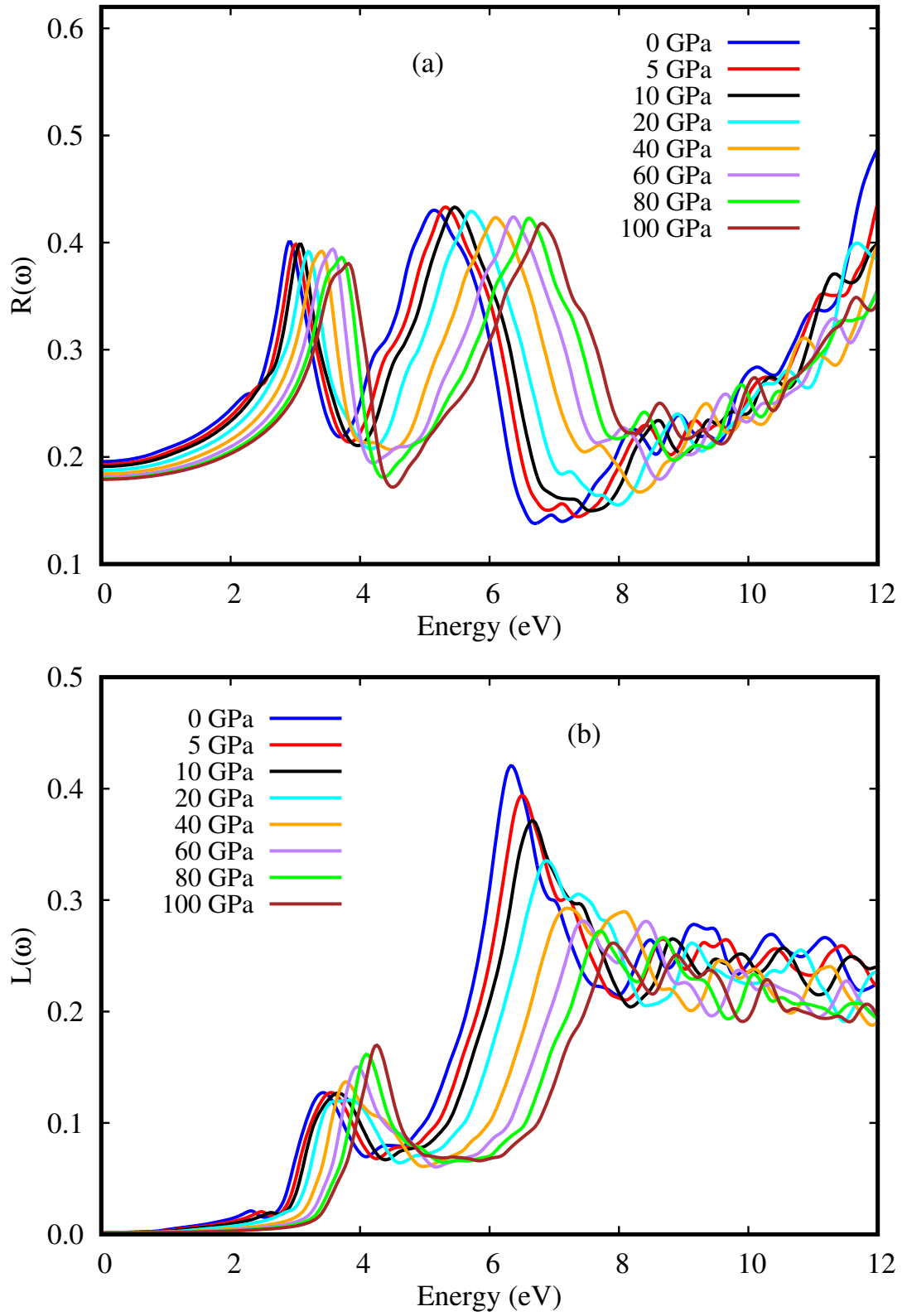


Figure 4.11: The pressure induced (a) Reflectivity $R(\omega)$ and (b) Energy loss $L(\omega)$ of $\text{Ba}_2\text{NbRhO}_6$ double perovskite as a function of photon energy.

The computed plot of $L(\omega)$ under varying hydrostatic pressures presented in Figure 4.11(b), reveals distinct characteristics in the visible and ultraviolet regions. In the visible range, the $L(\omega)$ shows lower peak compared to the UV region at all pressure condition. For energies exceeding 8.00 eV, the behavior becomes almost identical at all pressures. The peaks in this function correspond to the plasma resonance of valence electrons and are associated with the plasma frequency. A prominent peak of $L(\omega)$ is observed at 6.38 eV at 0 GPa, corresponding to a resonant energy loss in the UV range.

4.5 Mechanical Properties

Charpin's approach was used to investigate the mechanical properties of $\text{Ba}_2\text{NbRhO}_6$ double perovskite. The properties of a material play a vital role for engineering and designing applications that demand mechanical strength, durability, and stability. These properties includes elastic constants (C_{ij}), Young's modulus, Poisson's ratio, bulk modulus, shear modulus, Cauchy pressure, brittleness, ductility, and anisotropy. The mechanical stability of a cubic crystal structure can be evaluated using three independent elastic constants: C_{11} , C_{12} , and C_{44} . These constants are derived from a set of equations and must fulfill the Born stability criteria for the cubic system [75] ($C_{11} - C_{12} > 0$, $(C_{11} + 2C_{12}) > 0$, $C_{11} > 0$, $C_{44} > 0$ and $C_{12} < B < C_{11}$). The calculated elastic constants satisfy the mechanical stability conditions at all applied hydrostatic pressures are presented in Table 4.4. The findings show that C_{44} has the lowest value among the elastic constants, suggesting that the material is easily compressible in that direction. The Voigt–Reuss–Hill averaging method [86–88] was employed to estimate the elastic properties of the compound through the elastic constants C_{11} , C_{12} , and C_{44} . In this approach, the Voigt limit defines the bulk modulus (B_V) and the shear modulus (G_V) of a cubic crystal structure, which can be approximated as follows

$$B_V = \frac{(C_{11} + 2C_{12})}{3}. \quad (4.15)$$

$$G_V = \frac{1}{3}(3C_{44} + C_{11} - C_{12}). \quad (4.16)$$

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The Reuss bounds for the bulk modulus (B_R) and shear modulus (G_R) can be obtained by the following expressions

$$B_R = B_V. \quad (4.17)$$

$$G_R = \frac{5(C_{11} - C_{12})C_{44}}{4C_{44} + 3(C_{11} - C_{12})}. \quad (4.18)$$

Hill's elastic moduli approximation is employed to obtain the arithmetic mean of the Voigt and Reuss bounds.

$$B = \frac{B_V + B_R}{2}. \quad (4.19)$$

$$G = \frac{G_V + G_R}{2}. \quad (4.20)$$

Where, G_V and G_R are the bulk modulus of Voigt and Reuss. Young's modulus (Y), Pugh's ratio (B/G), Cauchy pressure (C_P), and Zener's anisotropy (A) [89] can be calculated as

$$Y = \frac{9GB}{3B + G}. \quad (4.21)$$

$$C_P = C_{12} - C_{44}. \quad (4.22)$$

$$A = \frac{2C_{44}}{(C_{11} - C_{12})}. \quad (4.23)$$

Young's modulus (Y) is a measure of a material's stiffness, with higher values indicating greater resistance to elastic deformation. Hence, it is expected that the calculated Young's modulus of the compound will illustrate its rigidity. Table 4.4 demonstrates that the stiffness of the studied material increases with increase in hydrostatic pressure because pressure forces the atoms closer, therefore, the compound exhibits maximum stiffness at 100 GPa. Cauchy pressure (C_P) characterizes not only the ductile or brittle nature of a material but also provides insight into the dominant type of atomic bonds. Materials with positive C_P are typically ductile and exhibit ionic bonds, while those with negative value indicates brittle nature with predominantly covalent bonds. The computed positive value of C_P confirms the material's ductility and implies greater dominance of ionic bonds under different

Table 4.4: Calculated result of elastic constants C_{11} (GPa), C_{12} (GPa), C_{44} (GPa), bulk modulus B (GPa), shear modulus G (GPa), Young's modulus Y (GPa), Cauchy's pressure C_P (GPa), Pugh ratio B/G , elastic anisotropy factor A (GPa), and Poisson's ratio ν of $\text{Ba}_2\text{NbRhO}_6$ double perovskite under different pressures.

Pressure (GPa)	C_{11} (GPa)	C_{12} (GPa)	C_{44} (GPa)	B (GPa)	G (GPa)	Y (GPa)	C_P (GPa)	B/G	A	ν
0	225.89	120.66	93.21	155.74	74.10	191.87	27.45	2.10	1.77	0.29
5	290.77	134.09	97.02	186.32	89.06	230.46	37.07	2.09	1.24	0.29
10	284.38	142.55	101.07	189.83	87.69	227.97	41.48	2.16	1.43	0.30
20	335.64	171.66	105.64	226.32	95.45	251.05	66.02	2.37	1.29	0.32
40	535.58	174.00	123.36	294.53	143.82	371.06	50.64	2.05	0.68	0.29
60	537.81	232.82	125.51	334.49	135.69	358.57	107.31	2.47	0.82	0.32
80	798.42	203.46	129.08	401.78	181.66	473.59	74.37	2.21	0.43	0.30
100	901.14	222.94	141.52	449.01	202.55	528.19	81.43	2.22	0.31	0.30

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hydrostatic pressures. The shear modulus (G), or modulus of rigidity, is a material's property that quantifies the resistance to transverse deformation, which is related to its hardness. Table 4.4 exhibits that the value of G rises with induced pressure that's why the material's hardness increases. Pugh introduced the concept of the ratio of bulk modulus to shear modulus (B/G) for polycrystalline materials. The intrinsic physical characteristics of a material, such as brittleness, ductility, directional properties, and stiffness, are governed by the B/G ratio and Poisson's ratio (ν). When the B/G ratio exceeds 1.75 and ν surpasses 0.26, the material typically exhibits ductile behavior, making it suitable for device fabrication otherwise brittle [90,91], and also categorized as plastic if it has a value equal to 0.26. It is evident from the B/G and ν values that the studied material is ductile under ambient and pressurized conditions. Anisotropy describes the variation of physical properties depending on direction in a material. In anisotropic materials, the properties are not the same in all directions. If the anisotropic ratio (A) is 1, the material is perfectly isotropic; otherwise it will be anisotropic [92]. The calculated value of A is 1.77 at ambient condition and decreases with increasing pressure, reaching 0.31 at 100 GPa, but the value of A is not equal to 1 throughout the pressure range, confirming that the material is anisotropic at all pressurized conditions.

4.6 Thermoelectric Properties

Thermoelectric materials have been recognized for their unique ability to convert waste heat into electrical energy and this capability have attracted the attention of researchers. BoltzTraP computational code was used to evaluate thermoelectric parameters such as electrical conductivity (σ), thermal conductivity (κ), Seebeck coefficient (S), power factor (PF), and figure of merit (ZT) of $\text{Ba}_2\text{NbRhO}_6$ double perovskite are estimated from 100 K to 1000 K, under different hydrostatic pressures ranging upto 100 GPa. BoltzTraP performs calculations based on the constant relaxation time approximation and evaluates σ , κ , and PF in terms of relaxation time (τ) for all electronic states.

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The relaxation time of charge carriers is given by following expression [93]

$$\tau = \frac{\hbar^3 C_{2D}}{k_B T m_d E_\beta^2} \quad (4.24)$$

where, C_{2D} denotes the two-dimensional elastic constant, k_B is the Boltzmann constant, E_β is the deformation potential constant of charge carriers along the β direction, m_d represents the effective mass of charge carriers and T is the temperature in Kelvin.

The electrical conductivity of a material is influenced by the mobility and density of its charge carriers. By analyzing how charges move, we can compute the behavior of a compound as conductor, semiconductor, or insulator. The electrical conductivity of double perovskite is crucial for their implementation in electronic devices such as sensors, energy conversion systems, and related technologies. The calculated electrical conductivity σ per relaxation time τ as a function of temperature for the studied material is depicted in Figure 4.12(a), under different hydrostatic pressures, which demonstrates that the conductivity increased linearly with both rising temperature and pressure, further confirming the semiconducting nature of the material. This material exhibits lower conductivity of σ/τ , specifically $2.55 \times 10^{19} (\Omega \text{ms})^{-1}$ at ambient pressure and 1000 K, while at 100 GPa it reaches a maximum value of $3.33 \times 10^{19} (\Omega \text{ms})^{-1}$. The increased value of σ/τ indicates the generation of more free electrons and the corresponding increase in carrier concentration. The high σ/τ of $\text{Ba}_2\text{NbRhO}_6$ double perovskite suggests that it has significant potential for applications in the thermoelectric industry.

Thermal conductivity is crucial in evaluating the thermoelectric applications of materials, and it describes how heat is transferred owing to the contributions of electrons and phonons. Thermal conductivity (κ/τ) is comprised of two components: (1) the electronic (κ_e/τ) contribution, which reflects the effect of electrons and holes within the material, and (2) the phonon lattice κ_l/τ contribution. This can be mathematically expressed as:

$$\frac{\kappa}{\tau} = \frac{\kappa_e}{\tau} + \frac{\kappa_l}{\tau}. \quad (4.25)$$

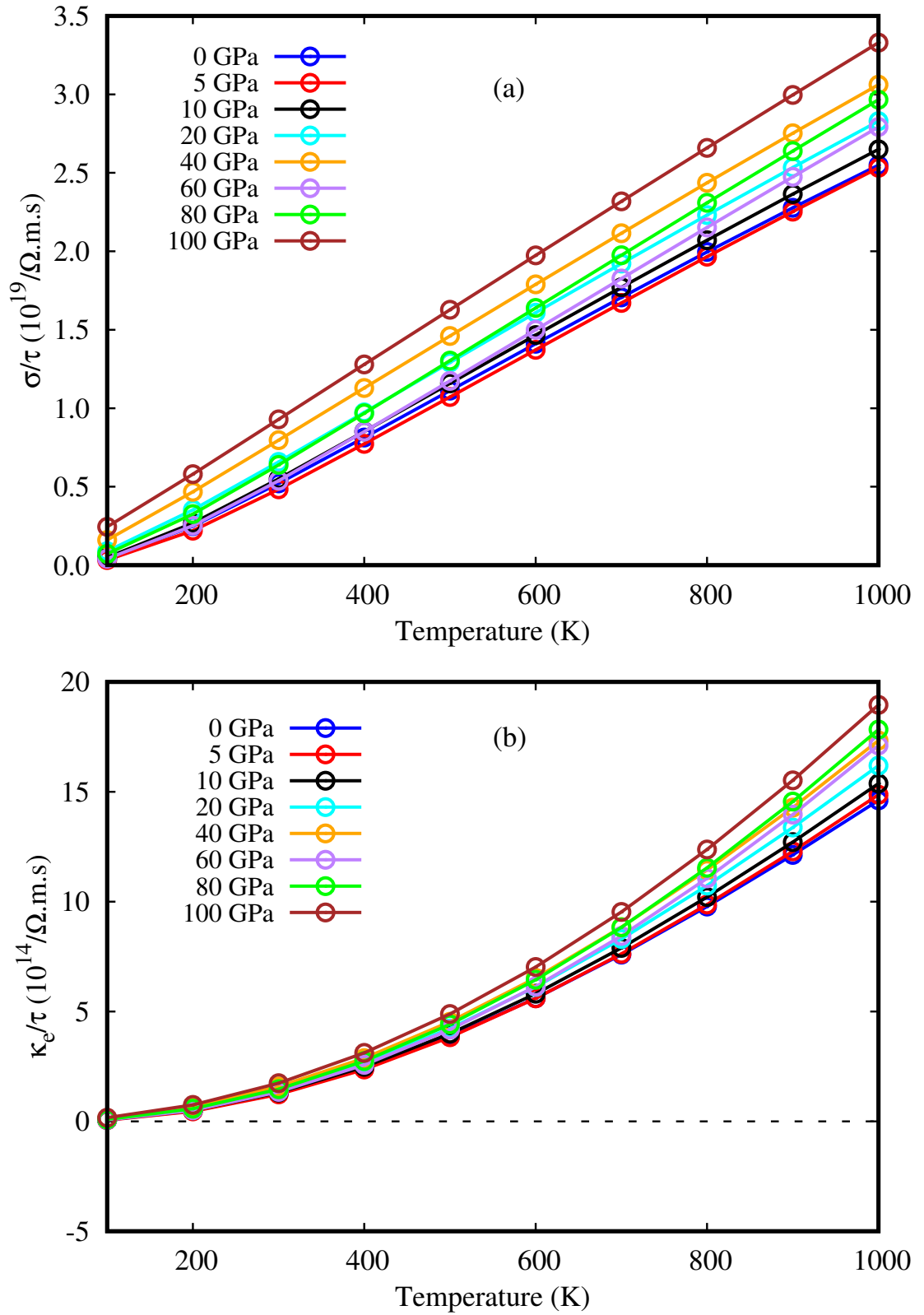


Figure 4.12: The calculated Thermoelectric properties of Ba₂NbRhO₆ double perovskite including (a) Electrical conductivity (σ/τ) and (b) Electronic thermal conductivity (κ_e/τ) under pressurized conditions.

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According to Slack's model [94] the temperature dependent lattice thermal conductivity (κ_l/τ) can be expressed as

$$\kappa_l = A \frac{M\theta_D^3 n^{\frac{1}{3}} \delta}{\gamma^2 T} = A \frac{M\theta^3 \delta}{\gamma^2 T n^{\frac{2}{3}}}. \quad (4.26)$$

since, $\theta_D = \theta n^{-\frac{1}{3}}$. Where, M , A , and δ represents the average atomic mass, the coefficient dependent on γ and the cube root of the average atomic volume, while n , T , θ_D , and γ are the number of atoms in the primitive cell, absolute temperature, Debye temperature, and Grüneisen parameter, respectively. The Grüneisen parameter (γ) [95] can be estimated using the Poisson's ratio as follows

$$\gamma = \frac{9(v_1^2 - \frac{4}{3}v_t^2)}{2(v_1^2 - 2v_t^2)} = \frac{3}{2} \left(\frac{1 + \nu}{2 - 3\nu} \right). \quad (4.27)$$

The coefficient A can be computed with the help of γ ,

$$A = \frac{2.43 \times 10^{-6}}{1 - \frac{0.514}{\gamma} + \frac{0.228}{\gamma^2}}. \quad (4.28)$$

The calculated values of A and γ are listed in Table 4.5. The electronic thermal conductivity (κ_e/τ) in Figure 4.12(b) displays a pattern similar to that of the electrical conductivity (σ/τ); however, its magnitude is considerably lower. To illustrate the performance utilized Wiedemann-Franz law, the ratio of κ_e and σ in the range of 10^6 suggests that the contribution of σ/τ is significantly larger than that of κ_e/τ . The value of κ_e/τ for $\text{Ba}_2\text{NbRhO}_6$ is of the order of 10^{14} , this reveals the suitability of this composition for thermoelectric device applications. The electronic portion of thermal conductivity under different hydrostatic pressures is illustrated in Figure 4.12(b), it demonstrates the value of κ_e/τ increases with the incremental application of pressure and also enhances with temperature. At 100 GPa, it exhibits significantly higher electronic thermal conductivity with a maximum value of 18.96×10^{14} (W/mKs) at 1000 K, whereas at pressures of 40 GPa and 80 GPa, its behavior remains almost identical. The rate of increase of κ_e/τ surpasses that of σ/τ with respect to both temperature and pressure. The calculated lattice thermal conductivity (κ_l/τ) plot of $\text{Ba}_2\text{NbRhO}_6$ under hydrostatic pressures are presented

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in Figure 4.13(a), it reveals the value of κ_l/τ reduce with increasing temperature but rises with the incremental application of pressure, consequently, at 100 K and 100 GPa, it illustrated the maximum value 96.46×10^{14} (W/mKs).

Table 4.5: Depicted values of average atomic mass (M), Debye temperature (θ_D), cubic root of average atomic volume (δ), Grüneisen parameter (γ), γ -dependent constant (A), and number of atoms per unit cell (n) for the studied compound at 300 K temperature under different hydrostatic pressures.

Pressures (GPa)	M ($\times 10^{-3}$ Kg/mol)	θ_D (K)	δ (Å)	γ	A	n
0	56.65	455.76	2.39	1.71	3.13	10
5	56.65	497.21	2.36	1.71	3.13	10
10	56.65	491.72	2.35	1.77	3.11	10
20	56.65	510.48	2.31	1.90	3.07	10
40	56.65	618.21	2.27	1.71	3.13	10
60	56.65	598.26	2.23	1.90	3.07	10
80	56.65	686.42	2.20	1.77	3.11	10
100	56.65	721.15	2.18	1.77	3.11	10

Seebeck coefficient measures (S) a material's ability to generate an electromotive force in response to an applied temperature gradient, in other words, it indicates the effectiveness of the thermocouple. To achieve high performance thermoelectric devices, a large Seebeck coefficient is required. It can be approximated by the formula [96]:

$$S = 8\pi^2 k_B m^* T (3eh^2)^{-1} \left(\frac{\pi}{3\rho} \right)^{\frac{2}{3}}. \quad (4.29)$$

Where, k_B , m^* , and e correspond to the Boltzmann constant, the effective mass of the charge carriers, and the elementary charge, respectively, whereas h , ρ , and T express the Planck's constant, the carrier concentration, and the temperature.

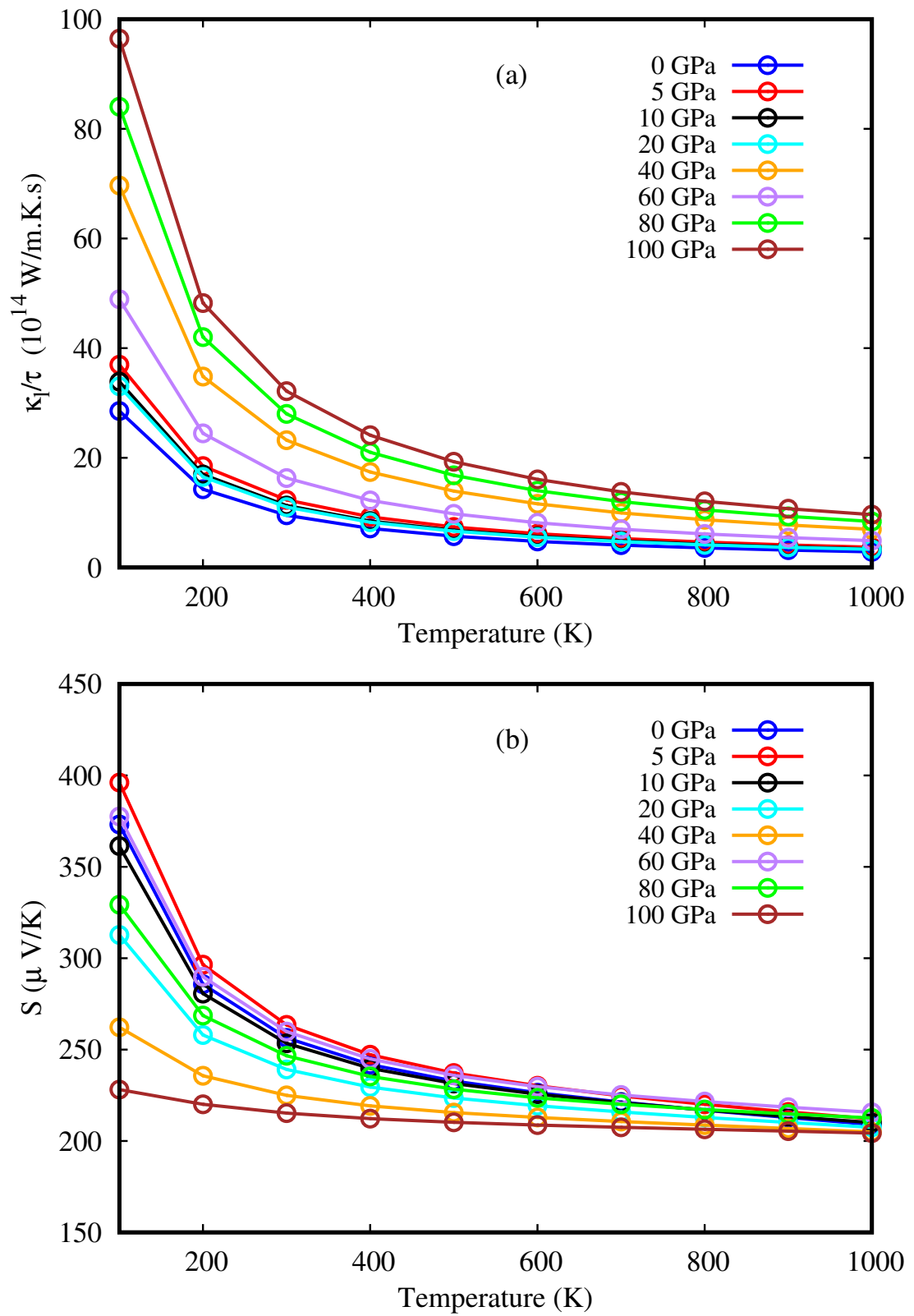


Figure 4.13: The calculated Thermoelectric properties of $\text{Ba}_2\text{NbRhO}_6$ double perovskite including (a) Lattice thermal conductivity (κ_l/τ) and (b) Seebeck coefficient (S) under pressurized conditions.

Results and Discussion

The graphical representation of S as a function of temperature under varying hydrostatic pressures is presented in Figure 4.13(b). It reveals a monotonous decline of S with increasing temperature across all pressures, although it does not attain negative values. At room temperature, Seebeck coefficient values equal to or greater than $200 \mu\text{V/K}$ are considered indicative of efficient thermoelectric performance [97]. The coefficient remains positive and stays consistently above this value for all applied pressures, which is shown in Table 4.6. Notably, the computed S reaches its maximum value for 5 GPa, observed across the temperature range from 100 K to 1000 K. Based on this investigation, it can be concluded that the compound $\text{Ba}_2\text{NbRhO}_6$ is a promising candidate for thermoelectric devices owing to its positive S across all pressures and it is particularly more suitable at pressures of 5 GPa and 60 GPa. Thermoelectric power factor (PF) is a dimensionless quantity, measures a material's efficiency in converting heat into electrical energy, and serves as a key measure of thermoelectric performance. Materials with a high power factor are preferred for effective thermoelectric applications. It can be expressed by the formula [98]

$$PF = \frac{S^2\sigma}{\tau}. \quad (4.30)$$

Enhancing the value of PF, we can improve the conversion efficiency of thermoelectric devices, which opens the door to its applications in waste heat recovery and portable power generation. Figure 4.14(a) exhibits the value of PF increases with induced pressure and also temperature. As a results, the calculated value of PF is maximum for 100 GPa at 1000 K and gives 209.01 ($\mu\text{V/K}$). The Figure of merit (ZT) defines the efficiency of a thermoelectric materials and is crucial for predicting the performance of the transport properties of compounds. To evaluate ZT, it is essential to consider the material's electrical conductivity (σ), Seebeck coefficient (S), and thermal conductivity (κ). The formula for ZT [99] can be expressed as follow

$$ZT = \frac{S^2\sigma T}{\kappa} = \frac{S^2\sigma T}{(\kappa_e + \kappa_l)}. \quad (4.31)$$

Figure of merit is directly proportional to both the Seebeck coefficient and the electrical conductivity; therefore, it increases with higher values of both parameters.

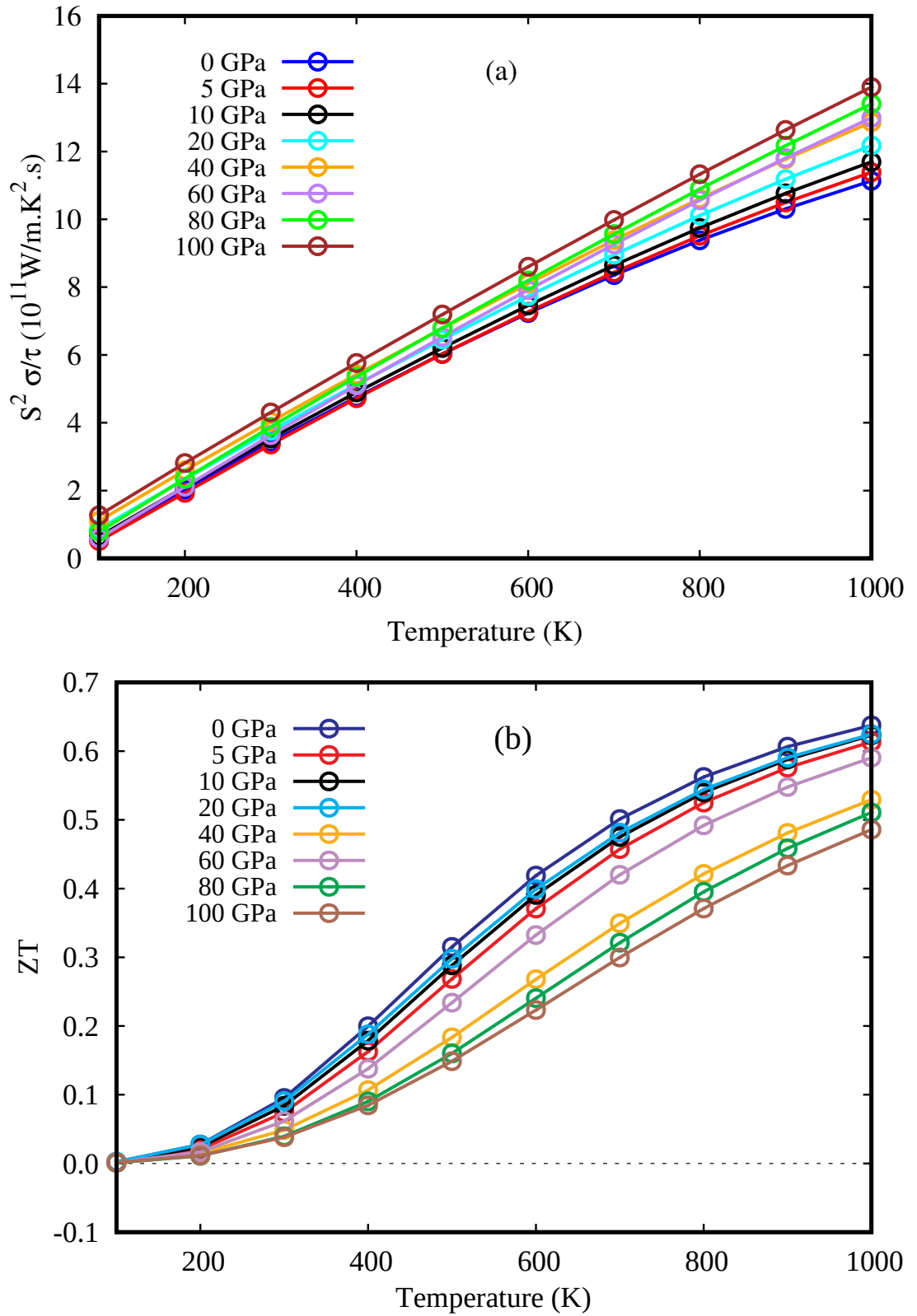


Figure 4.14: Thermoelectric properties of $\text{Ba}_2\text{NbRhO}_6$ double perovskite including (a) Power factor ($S^2 \sigma / \tau$) and (b) Figure of merit (ZT) under different hydrostatic pressures.

Results and Discussion

Table 4.6: Calculated values of electrical conductivity (σ/τ), electronic thermal conductivity (κ_e/τ), Seebeck coefficient (S), power factor (PF) and figure of merit (ZT) for the studied compound at 300 K temperature and different hydrostatic pressures.

Pressures (GPa)	σ/τ ($\times 10^{19}/\Omega\text{ms}$)	κ_e/τ ($\times 10^{14}$ W/mKs)	S ($\mu\text{V/K}$)	PF ($\times 10^{11}$ W/mK ² s)	ZT
0	0.52	1.27	256.42	3.43	0.10
5	0.48	1.23	263.54	3.35	0.07
10	0.55	1.31	253.43	3.53	0.08
20	0.66	1.44	239.14	3.76	0.09
40	0.79	1.59	224.95	4.03	0.05
60	0.54	1.34	259.89	3.64	0.06
80	0.63	1.46	246.59	3.88	0.04
100	0.93	1.74	215.25	4.30	0.04

Figure 4.14(b) represents the pressure-dependent behavior of the figure of merit (ZT) for Ba₂NbRhO₆ double perovskite from 100 K to 1000 K. The Figure illustrates that the ZT value increases progressively with temperature, while it exhibits a slight decrease with applied pressure. Consequently, the highest ZT value is obtained at ambient pressure and the lowest occurs at 100 GPa. At room temperature, ZT attains a maximum value of 0.10 at 0 GPa, decreases with increasing pressure, and eventually becomes constant after 80 GPa.

Conclusions

The FP-LAPW method within the framework of DFT was employed to study physical properties of double perovskite $\text{Ba}_2\text{NbRhO}_6$ under hydrostatic pressure ranging up to 100 GPa. For the most stable structure of the material, the estimated lattice constant is found to be 8.165 Å at ambient pressure, and it decreases gradually with increasing pressure. The studied compounds crystallize in a perfect cubic structure belonging to the space group $Fm\bar{3}m$ (#225). The negative values of formation energy confirm the thermodynamic stability. The electronic band structure of $\text{Ba}_2\text{NbRhO}_6$ exhibits semiconducting behavior with an indirect bandgap of 0.861 eV that increases to 1.860 eV at 100 GPa pressure. The pressure induced enhancement of the bandgap in $\text{Ba}_2\text{NbRhO}_6$ reveals its tunable optoelectronic behavior, indicating the material's potential for pressure dependent electronic modulation. With increasing pressure, the Ba-3d and Nb-3d orbitals move away from the Fermi level in the conduction band, contributing to an increase in the band gap. The effective mass ratio of $\text{Ba}_2\text{NbRhO}_6$ double perovskite exceeds unity over the entire pressure range, indicating the material's promising potential for photocatalytic applications. The computed density of states reveals that, at ambient pressure, electrons in oxygen atoms significantly contribute to the valence band, whereas at 20 GPa, they are found to predominantly contribute to the conduction band. The optical properties

Conclusions

analysis indicated that the material exhibits high optical absorption and conductivity in ultraviolet region for all applied pressure. The absorption spectra were calculated to investigate the material's capability to absorb light at different energy levels. These optical properties suggest that $\text{Ba}_2\text{NbRhO}_6$ is a promising candidate for applications in photovoltaics and optoelectronic devices under hydrostatic pressure. The analysis of Pugh's ratio, Cauchy pressure, and Poisson's ratio reveals that the studied material has ductile nature within the pressure range from 0 to 100 GPa. At 100 GPa, the compound shows significantly greater hardness compare to other pressure. The material exhibits elastic anisotropy, as indicated by anisotropy values which are greater and less than one at all hydrostatic pressure values. Furthermore, thermoelectric analysis reveals high electrical conductivity, positive Seebeck coefficient, and high figure of merit value. The positive values of Seebeck coefficient reveal the p-type semiconductor nature of the material. At room temperature, the value of Seebeck coefficient remains above $215 \mu\text{V}/\text{k}$ for all applied pressures. $\text{Ba}_2\text{NbRhO}_6$ has a figure of merit value of 0.64 at ambient pressure, with no significant change observed under applied pressure.

Therefore, $\text{Ba}_2\text{NbRhO}_6$ is regarded as an excellent candidate for optoelectronic, and thermoelectric applications owing to its promising optoelectronic and thermoelectric properties at ambient and pressurized conditions.

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